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Computer Science Research Report

# When Is a Time-Series Change Local? MetaShift-Bench for Selective Cross-Site Auditing


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| Student author(s) | *[Human completion required]* |
| School, province/state, country | *[Human completion required]* |
| Supervising teacher(s) | *[Human completion required]* |
| Submission date | *[Human completion required]* |


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# When Is a Time-Series Change Local? MetaShift-Bench for Selective Cross-Site Auditing

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## Abstract

A detector can recognize that a monitored time series changed without knowing whether the change is local to one target or shared across a comparison network. That distinction determines which follow-up evidence is relevant, yet real monitoring records rarely provide reliable scope labels. We formulate *selective cross-site auditing* as an information-constrained problem: at a stated scope-error tolerance, which cases can an observation channel answer, and when should it abstain?

MetaShift-Bench holds the target path exactly fixed across paired local and shared synthetic worlds and varies only donor-side participation. Its primary target-fixed experiment contains 360 held-out components, 230,400 matched pairs, and a complete $5 \times 4 \times (2^5) = 640$-cell adverse-condition grid: five donor-participation levels, four signal strengths, and five binary stress factors. In the finite predeclared policy set, target-only information has no positive held-out coverage below 50% tolerated error. Comparative information also answers none through $\alpha = .10$, then reaches 60.4% coverage at $\alpha = .20$ with 19.5% observed conditional error. A structural interval-separation certificate answered 89,997 of 230,400 matched pairs (39.1%), with no observed errors among 179,994 answered scope arms. It recovered 97.7% of the pairs in a predeclared simulation-information reference region and abstained on the remaining 140,403 pairs (60.9%), including all 46,080 observationally indistinguishable $q = 0$ negative controls. These finite-grid results hold only under the bounded synthetic generator and do not constitute a population-risk guarantee or a deployable real-network certificate.

We prove the target-only impossibility statement under matched, balanced target laws, derive the availability-normalized partial-scope residual gap, and give a sufficient condition for structural abstention. A separate AQS deployment audit measures comparison availability and abstention in real public records; it supplies neither scope labels nor mechanism truth. In an independent stable-regime benchmark, cross-site methods attain AUPRC 0.884–0.915, while target-only baselines remain at 0.500–0.501. MetaShift has no confidence-supported aggregate advantage over standard synthetic control. The central contribution is therefore not estimator superiority, but an evaluation framework in which an audit answers a scope question only when its comparison evidence can support that answer.

**Keywords:** local-versus-shared change; time-series auditing; selective prediction; cross-site comparison; structural abstention.

# Contents

# 1 Introduction

A reported $PM_{2.5}$ Method Code transition can coincide with an abrupt change at one monitor. An isolated change directs attention toward target-specific records, whereas simultaneous changes at nearby monitors call for a network-wide explanation. A target-only detector can identify the numerical break, but it cannot determine whether its scope is local to the target or shared across the comparison network. Forcing a scope label when the necessary comparison is unavailable can misdirect a technical investigation just as readily as missing a change.

MetaShift-Bench frames this as selective prediction over explicitly limited time-series information. The central question is not "which estimator wins?" It is:

> *Given a specified observation channel and an allowed scope-error rate, what fraction of scope questions can a system answer, and when must it abstain?*

We call this *information-constrained selective scope auditing.* Here, *scope* means whether a change is specific to the target or shared by its declared comparison network. It must be separated from detection and mechanism (Table 1).

The same information structure appears whenever a focal stream is compared with a reference network, including sensor operations, platform telemetry, and climate-record homogenization. This report evaluates neither those deployments nor their mechanisms; it studies the transferable question of what a specified observation channel can justify.

**Table 1:** Three distinct questions raised by a changing monitored time series.

| Question | What it asks | Minimum evidence needed |
|---|---|---|
| Detection | Did the target path change? | Target time series. |
| Scope | Is the change local or shared with the comparison network? | Target series plus a qualified cross-site comparison. |
| Mechanism | Why did the change occur? | Technical records, domain metadata, and human review in addition to a comparison. |

A reported AQS Method Code transition is useful as an observable metadata anchor, but it is not evidence that a device was replaced, failed, or biased. That distinction makes the real-data audit a test of comparison availability and principled abstention, rather than a source of unobserved scope or mechanism labels.

A scope decision has practical consequences even when its mechanism remains unknown: it determines whether an investigator should prioritize target-specific records or a network-wide explanation. An unsupported local/shared label can therefore send a limited technical review toward the wrong evidence.

MetaShift-Bench supplies a deliberately target-fixed synthetic test of the scope question. Every matched pair has exactly identical target observations in the local and shared arms; only donor participation changes. Consequently, a target-only policy cannot distinguish balanced local from shared labels in the stipulated experiment, while a policy receiving target and donor channels may do so. The benchmark measures the held-out coverage achieved by predeclared policies at stated answered-case error tolerances.

Figure 1 gives a compact constructed example. The target trace is identical in the two arms; only the donor response, and therefore the target–donor residual, distinguishes local from shared scope.

The primary design extends a binary local-versus-regional endpoint into a donor-participation continuum. It crosses nominal participation $q \in \{0, .25, .50, .75, 1\}$ with signal strength and

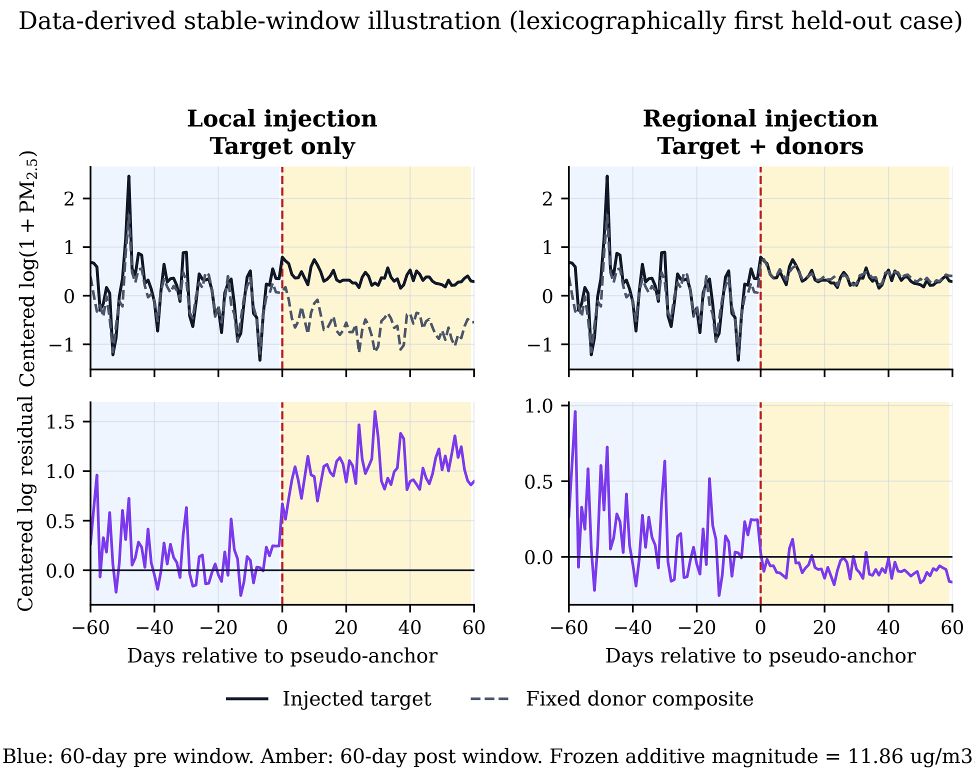


**Figure 1:** A deterministic stable-regime synthetic illustration of the target-fixed paired construction. The injected target effect is identical in both arms: the local arm changes the target only, whereas the regional arm also changes the donors. This constructed comparison is not physical-mechanism evidence from the underlying AQS paths.

adverse comparison conditions. A bounded structural certificate answers only when availability-normalized score intervals separate under declared generator envelopes. A nonpositive margin is retained as a required abstention region, not hidden inside a weak classifier score.

The report asks five linked research questions:

1. **RQ1:** How much scope-answerability coverage do target-only and comparative channels achieve at prespecified error tolerances?
2. **RQ2:** When does a structural certificate answer or abstain along the partial-scope boundary?
3. **RQ3:** Can cross-site methods separate the constructed scope task from single-series methods, and is any MetaShift advantage over standard synthetic control supported?
4. **RQ4:** How often can a qualified cross-site audit be formed in the AQS deployment inventory, and what dispositions result?
5. **RQ5:** What do interval calibration and contextual evidence reveal about the limits of real-data auditing?

Our contributions are fourfold:

- **C1. Introduce a new evaluation problem:** selective scope answerability by observation channel, rather than unconditional scope classification or estimator ranking.
- **C2. Construct paired information worlds** that hold every target observation fixed while varying only donor-side scope information, including observationally identical negative controls.
- **C3. Derive construction-specific theory:** an availability-normalized partial-scope residual identity and a bounded interval-separation condition for structural abstention.
- **C4. Audit the evaluation end to end** with predeclared policies, a complete adverse-condition grid, retained failure cells, and an independent AQS availability inventory.

The framework builds on established selective classification, information refinement, change-point detection, and synthetic control. Its contribution is their bounded joint evaluation for an explicitly target-fixed cross-site scope question.

**Result preview.** In the target-fixed held-out construction, no predeclared target-only policy has positive finite-policy coverage below one-half tolerated scope error. Comparative policies reach positive coverage only at the least strict reported tolerance, $\alpha = .20$. The structural certificate abstains on most pairs and is not a real-network deployment policy. In the independent stable-regime benchmark, cross-site methods outscore target-only baselines on constructed scope separation, but Standard SC remains a strong baseline and neither MetaShift variant has a confidence-supported aggregate advantage. The AQS inventory reports comparison availability and diagnostic evidence tiers, not physical monitor mechanisms. These findings motivate evaluating whether a scope answer is justified, not merely whether a detector predicts one.

# 2 Problem formulation

## 2.1 Three questions and observation channels

For one target time series, a target-only channel sees the target path itself; a comparative channel adds donor paths and their availability; an external channel would additionally contain admissible station or maintenance records. Let $Y \in \{L, S\}$ denote a stipulated synthetic scope label, where $L$ is target-local and $S$ is shared with comparative donors. The conceptual nested channels are

$$O_T = X, \qquad O_C = (X, D), \qquad O_E = (X, D, E).$$

Here $X$ is the declared target path, $D$ contains donor paths, fixed donor weights, and availability indicators, and $E$ denotes admissible external records. The synthetic experiment supplies constructed labels only for detection and scope. It contains neither an empirical $O_E$ channel nor mechanism labels.

A selective policy $\phi = (g, s)$ maps an allowed observation channel and auxiliary randomness to a scope prediction $g$ and an answer indicator $s \in \{0, 1\}$. Its population answerability frontier at tolerated conditional scope error $\alpha$ is

$$\Gamma_{\mathcal{I}}(\alpha) = \sup_{\substack{\phi\ \mathcal{I}\text{-measurable} \\ \Pr(s_\phi = 1) > 0 \\ \Pr(g_\phi(O) \neq Y \mid s_\phi = 1) \leq \alpha}} \Pr(s_\phi = 1).$$

The supremum is defined as zero if no positive-coverage policy meets the constraint. This is an inverse selective risk–coverage objective specialized to a scope label; it is not a new generic risk–coverage definition [22, 23].

The tolerance $\alpha$ is the largest permitted error rate among cases that a policy answers. Abstained cases do not enter this conditional error. In the experiment, the theoretical frontier

**Table 2:** Notation for the target-fixed partial-scope construction.

| Symbol | Meaning |
|---|---|
| $a_{jt}$ and $\widetilde{w}_{jt}$ | Donor $j$'s availability and its availability-normalized weight on date $t$. |
| $\lambda_{jt}$ and $q_t$ | Declared donor participation and realized weighted participation on date $t$. |
| $h_t$ and $H$ | Injected target schedule and its constant-schedule magnitude. |
| $A$ | Post-minus-pre mean residual score used for the scope decision. |
| $B_L, B_S, \kappa$ | Declared local/shared error envelopes and structural separation margin. |

is not directly observed. Instead, a finite set of predeclared policies is evaluated on held-out components.

For nested channels, the *Scope Answerability Gain* is

$$\Delta_{\mathcal{I}\to\mathcal{J}}(\alpha) = \mathrm{SAG}_{\mathcal{I}\to\mathcal{J}}(\alpha) = \Gamma_{\mathcal{J}}(\alpha) - \Gamma_{\mathcal{I}}(\alpha).$$

The benchmark-specific name does not make its nonnegativity a new decision-theoretic principle. If the richer channel can discard its additional coordinates, its policy set includes the poorer-channel policy set, a Blackwell-style information-refinement argument [14, 15].

## 2.2 Target-only impossibility and finite-policy estimation

**Proposition 1 (target-fixed impossibility).** Suppose the synthetic labels are balanced and the matched construction satisfies

$$\Pr(Y = L) = \Pr(Y = S) = \frac{1}{2}, \qquad \mathcal{L}(X \mid Y = L) = \mathcal{L}(X \mid Y = S).$$

If selection and prediction use only $X$ and auxiliary randomness that is conditionally independent of $Y$ given $X$, then every positive-coverage target-only policy has conditional scope error $1/2$. Therefore

$$\Gamma_T(\alpha) = 0 \qquad \text{for } \alpha < \frac{1}{2}.$$

*Proof.* Conditioning on a target-only answer does not change the balanced posterior label probability because the accepted target law is the same under both labels. Either binary prediction is consequently wrong with probability $1/2$. □

This is not a generic impossibility theorem for time series. It is the correctness argument for the benchmark's matched construction: it would fail if target-only input leaked a pair identifier, donor availability, participation value, label schedule, generator truth, or label-dependent randomness.

**Observation 1 (information refinement).** When $O_C$ contains $O_T$ under the same joint law, label space, loss, and policy class,

$$\Gamma_C(\alpha) \geq \Gamma_T(\alpha).$$

The result follows by implementing every target-only policy after discarding $D$. It neither implies strict improvement nor validates any finite-sample comparative classifier.

The experiment estimates neither unrestricted population frontier. It reports the held-out finite-policy envelope

$$\widehat{\Gamma}^{\mathcal{P}}_{\mathcal{I}}(\alpha) = \max_{\substack{\phi\in\mathcal{P} \\ \widehat{R}_\phi \leq \alpha}} \widehat{C}_\phi,$$

with zero when no positive-coverage frozen policy qualifies. Here $\mathcal{P}$ contains the target-only forced rule, comparative forced rule, four separately calibration-selected confidence rules, and the specified synthetic-design-assisted certificate. Thus every displayed result is a descriptive held-out envelope for the frozen generator and policy set, not a population optimum, conditional-risk guarantee, or post-evaluation policy choice.

## 2.3 Target-fixed partial-scope construction

The primary construction is performed on the analysis scale $z = f(y) = \log(1 + \max(y, 0))$. Let $w_j \geq 0$ be fixed donor weights and $a_{jt}$ denote donor availability. On every retained date,

$$\widetilde{w}_{jt} = \frac{w_j a_{jt}}{\sum_k w_k a_{kt}}, \qquad q_t = \sum_j \widetilde{w}_{jt}\lambda_{jt},$$

where $\lambda_{jt} \in [0, 1]$ is declared donor participation. For positive post-anchor schedule $h_t$, the paired worlds are

$$z_t^L = z_t + h_t, \quad z_{jt}^L = z_{jt}, \qquad z_t^S = z_t + h_t, \quad z_{jt}^S = z_{jt} + \lambda_{jt} h_t.$$

The target path is exactly identical in both worlds. With fixed offset $b$,

$$r_t = z_t - \sum_j \widetilde{w}_{jt} z_{jt} - b, \qquad A = \mathrm{mean}_{W_{\mathrm{post}}} r_t - \mathrm{mean}_{W_{\mathrm{pre}}} r_t.$$

**Proposition 2 (construction-specific residual identity).** With arm-invariant availability, retained dates, weights, preprocessing, and pre-anchor offset,

$$r_t^L - r_t = h_t, \qquad r_t^S - r_t = (1 - q_t) h_t.$$

Intuitively, a local injection remains fully in the target residual, while a shared injection is partly canceled by donors. The residual gap is therefore realized donor participation times the injected signal. For a constant post schedule $h_t = H$, the score-center separation is

$$A^L - A^S = \overline{q_t H}.$$

For a variable schedule the correct expression is $\overline{q_t h_t}$, not $\bar{q}\,\bar{h}$. This is exact algebra for the stipulated analysis-scale construction, not a generic identifiability claim. A raw-scale additive field does not preserve this identity after the nonlinear transformation and is treated only through a conservative leakage envelope.

### 2.4 Structural Answerability Certificate

**The certificate is a synthetic-design-information-assisted policy, not a deployable real-network rule.** Denoted $O_C^+$, it receives generated participation and bounded-error quantities unavailable to a deployed monitoring system. For its efficiency diagnostic alone, the predeclared simulation-information reference region applies the same interval rule after replacing the conservative lower gap $G_{\min}$ with the generator-known realized gap $G = \overline{q_t h_t}$. It is an accounting reference, not a theoretical optimum, population upper bound, or deployment channel. Suppose its local and shared scores obey

$$|A^L - H| \le B_L, \qquad |A^S - (H - G)| \le B_S,$$

and all scored retained dates satisfy $q_t \ge q_{\min}$ and $h_t \ge H_{\min} > 0$. Let

$$G_{\min} = q_{\min} H_{\min}, \qquad \kappa = G_{\min} - (B_L + B_S).$$

**Proposition 3 (interval-separation certificate).** If $\kappa > 0$, the local and shared score intervals are disjoint. Every threshold in

$$(H - G_{\min} + B_S,\ H - B_L)$$

separates them under the stated bounds. The implementation uses their midpoint,

$$\tau_{\mathrm{cert}} = H - \frac{G_{\min}}{2} + \frac{B_S - B_L}{2},$$

answers only if $\kappa > 0$, and predicts local if $A > \tau_{\mathrm{cert}}$.

*Proof.* The upper shared bound is $H - G_{\min} + B_S$ and the lower local bound is $H - B_L$; their strict ordering is exactly $\kappa > 0$. □

The theorem requires bounded noise, fixed retained-date semantics, valid availability normalization, modeled donor mismatch, bounded contamination, and a clipping-aware raw-field bound. Nonfinite values, changed dates, unmodeled transformations, fewer than three donors, or an envelope violation invalidate the certificate. When $\kappa \leq 0$, it must abstain. A nominal midpoint $H - G_{\min}/2$ is not generally safe under asymmetric bounds; the asymmetric term above is necessary.

# 3 Related work

## 3.1 Information and selective prediction

Blackwell's comparison of statistical experiments formalizes when one observation can reproduce another decision problem by discarding or garbling information [14, 15]. The nested-channel observation used here is a direct specialization: a comparative policy can always discard donor information and imitate a target-only policy. It does not imply that any fitted rule improves in a finite sample or that comparative records are valid in an operational network.

Selective classification and reject-option learning already study the risk–coverage trade-off, optimal coverage at a chosen risk, and the cost of abstention [10, 18, 22, 23, 27]. Hierarchical selective classification permits a less-specific answer when a fine label is unjustified [29]. Conformal and predictive-inference work clarifies which assumptions are needed before interval statements are calibrated [6, 7, 35, 39, 48]. MetaShift-Bench uses these established ideas for a different selected object: a local-versus-shared scope judgment under explicitly contrasting observation channels.

## 3.2 Local/shared changes, residual signatures, and weak labels

Multivariate change-point work separates common and idiosyncratic components or tests common breaks in panels [5, 8]. Single-series procedures such as CUSUM, PELT, and multiscale segmentation remain useful for detecting a break, but cannot alone establish whether it is target-local or shared [25, 32, 41]. Weak-supervision research likewise distinguishes proxy labels from known truth [17, 24, 36, 38]. The synthetic scope labels in this report are defined by construction; a reported AQS Method Code is only a metadata anchor.

Residual signatures, analytical redundancy, fault distinguishability, and sensor placement are established in model-based diagnosis [19, 33, 40]. Climate-record homogenization similarly uses pairwise references and station histories to identify inhomogeneities [37, 49]. These literatures motivate the comparison logic, but their physical models and documented mechanisms are not assumed by this benchmark.

## 3.3 Counterfactual monitoring and public metadata

Synthetic control supplies donor-weighted counterfactuals and in-space placebo logic [1–3]. Generalized, augmented, and synthetic difference-in-differences variants address other panel structures and regularization choices [4, 11, 50]; modern difference-in-differences also emphasizes assumptions and treatment-timing heterogeneity [16, 28]. Moving-block bootstrap methods address serial dependence in time-series resampling [34], while multiple-testing and post-selection literature warns against overinterpreting screened findings [12, 13].

EPA publishes AQS file formats, Method Code and POC references, and monitoring comparability resources [43–47]. Collocated-monitor studies show why configuration-specific comparison matters [31]. Meteorological-normalization and low-cost sensor studies give additional context for

**Table 3:** Research traditions that inform the benchmark and the role retained here.

| Research tradition | Established object | Role in MetaShift-Bench |
|---|---|---|
| Selective prediction [18, 27] | Risk–coverage and reject-option policies. | Measures answered-case coverage for a scope question; does not rename the generic objective. |
| Common/idiosyncratic change analysis [5, 8] | Shared and unit-specific movement in multivariate series. | Holds the target fixed while varying donor participation in paired worlds. |
| Synthetic control and DiD [2, 4] | Donor-weighted comparisons under explicit assumptions. | Uses pre-anchor fitting and falsification diagnostics, not policy-style causal identification. |
| Residual diagnosis and homogenization [19, 37] | Signatures, reference series, and documented records. | Uses a bounded synthetic separation condition and keeps mechanism interpretation external. |
| Monitoring metadata [44, 47] | Administrative fields and technical documentation. | Uses Method Code as a reproducible audit anchor, not physical ground truth. |

air-quality comparison [9, 20, 21, 26, 30, 51]. They do not provide dated, site-level physical-change labels for the public AQS metadata anchors used here.

**Bounded gap.** The originality claim is deliberately narrow. Among the reviewed literature, we found no benchmark that jointly fixes the target path, varies donor participation, evaluates selective risk–coverage by observation channel, and requires structural abstention when comparison evidence is insufficient. This is a bounded literature statement, not a claim of universal priority. The contribution is the target-fixed evaluation construction, its availability-normalized residual identity, and its audited abstention boundary.

# 4 Synthetic benchmark construction

## 4.1 Target-fixed paired worlds

The primary benchmark is a constructed experiment, not a model of $PM_{2.5}$ transport or monitoring hardware. It generates bounded target-and-donor panels in memory from independent seed streams. Component, target, donor, and date identifiers have no physical-site identity and no relation to AQS data. This deliberate separation lets the experiment isolate one question: what additional scope information is carried by a comparison channel after the target observation has been held fixed?

Each matched pair contains a local arm and a shared arm with identical target arrays, target noise, base panel, and non-scope stress conditions. The arms differ only in the fraction of the injected target schedule received by available donors. At $q = 0$, donor arrays are also identical, creating an observationally indistinguishable negative control. At $q = 1$, all available donors receive the schedule. Intermediate values implement partial sharing after availability-normalized weights are applied on each retained date.

## 4.2 Predeclared stress grid

For every component, the complete $5 \times 4 \times (2^5) = 640$ grid crosses five nominal donor-participation values and four signal strengths with five binary stress factors: donor mismatch, availability loss, one-donor contamination, nonlinear raw-scale leakage, and bounded noise. The raw-scale field is an adversarial transformation test; it is not a claim about the frequency of such processes in an operational monitoring network. The construction requires exact target identity, complete grid cells, disjoint calibration and evaluation component sets, and no outcome-dependent configuration changes.

### 4.3 Relation to the AQS audit

The synthetic benchmark supplies constructed scope labels and therefore supports controlled scope-answerability evaluation. The AQS audit in Section 5 serves a different purpose: it measures whether a qualified comparison can be formed in public records and whether the protocol must abstain. It neither supplies labels for the synthetic experiment nor validates a real measurement mechanism.

## 5 AQS deployment audit data

The AQS audit is included to study deployment boundaries: whether public records support a qualified cross-site comparison and when the audit must abstain. It does not label local-versus-shared scope and does not identify a physical monitoring mechanism.

### 5.1 Primary public-data corpus

The primary corpus comprises seven annual EPA AirData daily archives for parameter 88101, $PM_{2.5}$ local conditions, spanning 2019–2025 [43, 45]. The snapshot was acquired on 2026-08-30 UTC. A versioned data gate records each official URL, archive checksum, selected CSV member, and retained row count, enabling reconstruction without placing large raw archives or API responses in the repository.

The canonical signal is the reported `Arithmetic Mean` for `24-HR BLK AVG` observations. The predeclared cleaning rule keeps finite values with observation percent at least 75, a non-excluded Event Type, and a nonempty Method Code. It then refuses duplicate records with the same state, county, site, POC, and date key. Thus alternative daily summaries cannot be treated as independent observations. The retained primary inventory is a frozen snapshot; it is not a claim that the public source is immutable or complete for all later AQS updates.

**Table 4:** Frozen v0.3.2 data and audit inventory.

| Quantity | Count |
|---|---:|
| Canonical daily PM2.5 records | 2,424,793 |
| Monitor time series | 1,689 |
| Stable pseudo-anchors from Method Code regimes | 124 |
| Stable pseudo-anchors from all-monitor regimes | 22 |
| Persistent Method Code anchors | 563 |
| Anchors with at least one distinct physical donor | 394 |
| Anchors with at least three distinct physical donors | 238 |
| Complete common-method comparisons | 228 |
| Donor-insufficient anchors | 325 |
| Estimator input-window failures | 10 |

*Note.* The physical donor identifier is State Code + County Code + Site Number; at most one POC is retained per donor site.

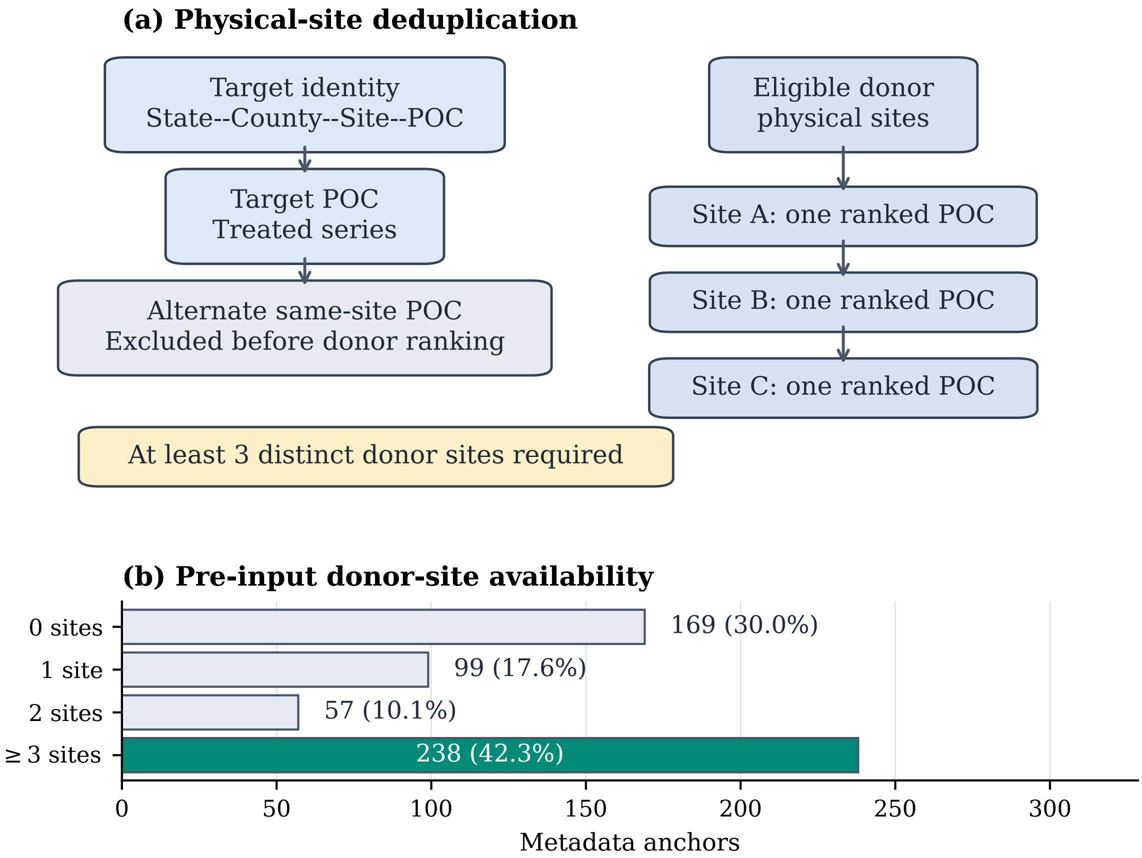


**Figure 2:** Physical-site donor construction and pre-input availability in the frozen 88101 inventory (563 metadata anchors). A target identity includes State–County–Site–POC; alternate POCs at the target physical site are excluded, and at most one ranked POC is retained at each other physical site. Only 238 of 563 anchors reach the three-distinct-site threshold before the common input-window check.

The frozen deployment inventory contains 563 persistent anchors, of which 238 pass the distinct-physical-donor eligibility screen before the separate common-input-window check. These counts describe where the legacy observational workflow can be formed; they are not labels for local scope or physical mechanism.

The daily data are public, but raw archives and API responses are not included in this manuscript repository or its paper assets. The display-only case-study renderer verifies archive checksums locally before reading the same source files. That separation supplies a reproducible source chain without placing raw data, credentials, or local cache files under version control.

## 5.2 Metadata anchors and distinct physical donors

An anchor is a persistent reported Method Code transition with at least 60 calendar days on each side, at least 45 retained observations in each adjacent method run, and no more than a seven-day transition gap. Its fields preserve the old and new reported codes and method names, but neither field establishes why the record changed. EPA network-data-alignment documentation supplies method context but does not date or explain an individual site transition [42]. An eligible geographic donor must be at a different physical site, within the predeclared 100-km radius, method-stable over the target window, observed on at least 60 paired pre-event days, and correlated at least 0.60 on the log scale before the anchor.

Physical donor identity is State Code + County Code + Site Number. A target's candidate donor list can contain multiple POCs for one physical site, but the ranked donor rule retains at most one: highest pre-event correlation, then shorter distance, then more paired days, then POC as a deterministic tie-breaker. An earlier internal inventory could count multiple POCs from one site as independent geographic donors. Every AQS result reported here uses the corrected physical-site rule; the correction does not affect the separate target-fixed benchmark, which has no AQS inputs. Same-site alternate POCs are reserved for contextual consistency analysis.

### 5.3 Stable-regime benchmark inputs

Real anchors do not provide known physical-bias truth. The benchmark therefore constructs pseudo-anchors only inside stable target and donor method regimes, at least 60 days away from a reported target or donor transition. The frozen case manifest contains 124 method-transition stable regimes and 22 all-monitor stable regimes. Both sources are retained because the target is a known, stable measurement record at the pseudo-anchor; neither is used as a real-event causal label. Section 7 explains how the target-plus-donor footprints are split before any held-out metric is calculated.

### 5.4 Audit disposition and parameter separation

Every anchor is retained in the event audit. A complete comparison means that the required common cross-site inputs exist; donor insufficiency and input-window failure are explicit machine-readable outcomes rather than omitted rows. This complete accounting is necessary because a result on only the well-matched subset would overstate the scope of cross-site inference.

Parameter 88502 is processed through a separate data gate, anchor inventory, and audit table. It is neither appended to nor pooled with the primary 88101 results. This separation prevents an apparent replication from being created by mixing parameter-specific reporting conventions, availability, or eligible event populations.

## 6 Selective audit methods

### 6.1 Policies for a specified information channel

Section 4 defines paired worlds; the policies here define what an auditor may do with each allowed channel. The target-only forced policy receives a signed target-change score and always predicts local. The comparative forced policy receives a target-minus-availability-normalized-donor residual score and uses a calibration-selected threshold. Four confidence-selective policies use the same comparative prediction with separate, calibration-chosen cutoffs for $\alpha = .01, .05, .10, .20$. The certificate policy is different: it answers only when the structural margin $\kappa$ in Section 2 is strictly positive.

### 6.2 AQS comparison workflow

The independent AQS audit follows a parallel but more limited workflow. Shared preprocessing turns public archives into canonical daily series and metadata anchors. Stable-window synthetic cases have constructed truth; observational anchors instead produce an audit disposition and diagnostic context. This separation prevents the real-data metadata from becoming an unobserved scope label.

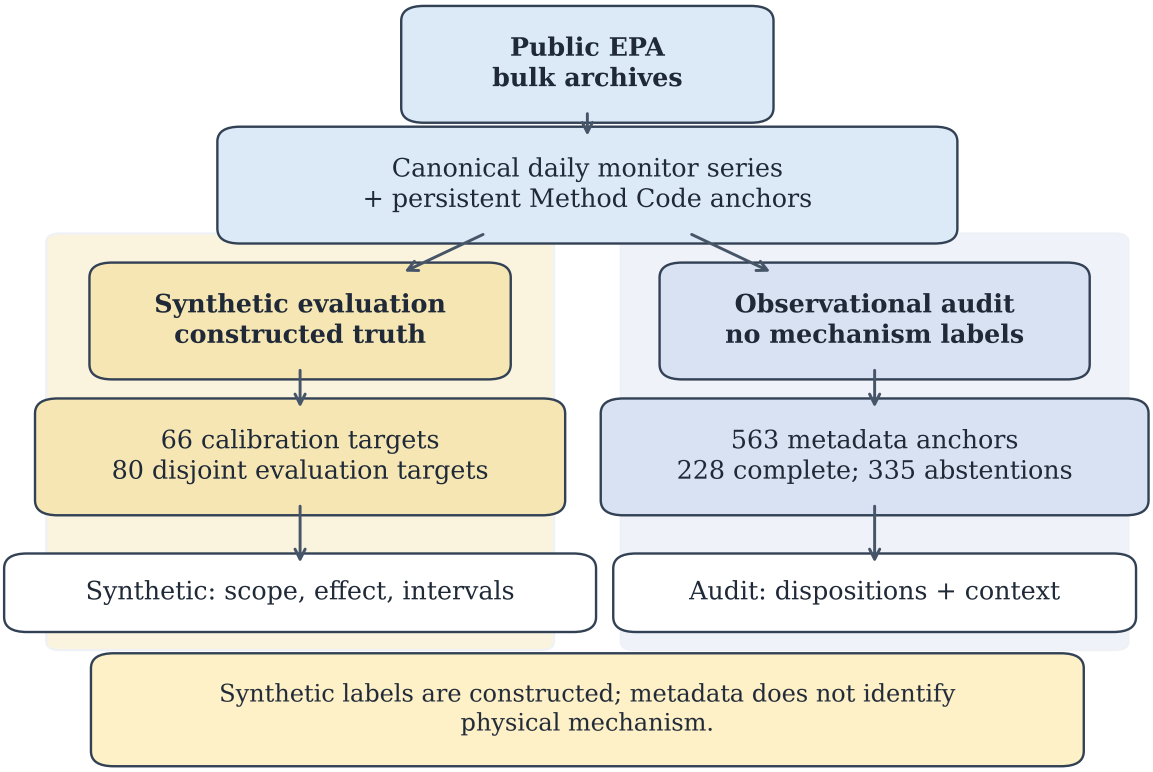


**Figure 3:** AQS deployment-audit workflow. Shared preprocessing produces canonical daily series and metadata anchors, then feeds a stable-window synthetic lane with constructed labels and an observational lane with audit dispositions. The observational lane supplies diagnostic context, not mechanism labels.

**Table 5:** Method roles in the independent stable-regime benchmark and AQS deployment audit.

| Method | Main input and fit | Metadata prior | Role |
|---|---|---|---|
| Standard synthetic control | Target and donor paths; nonnegative pre-anchor fit. | None. | Cross-site reference method in benchmark and AQS audit. |
| Fixed MetaShift | Target and donor paths; regularized pre-anchor fit. | Correlation–distance reliability prior. | Cross-site comparator in benchmark and AQS audit. |
| Cross-validated MetaShift | Same fit family with pre-period penalty selection. | Correlation–distance reliability prior. | Benchmark-only comparator; not introduced into the AQS audit after inspection. |
| Nearest-neighbor DiD | Target and highest-ranked donor. | Donor ranking only. | Cross-site benchmark comparator. |
| Single-series baselines | Target path alone. | None. | Detection contrast; no qualified regional counterfactual. |

## 6.3 Cross-site comparison methods

For a complete anchor, the donor pool is ranked by pre-event correlation, distance, paired days, and POC after physical-site deduplication. Standard synthetic control solves the nonnegative simplex problem

$$\min_{\substack{w_j \geq 0 \\ \sum_j w_j = 1}} \frac{1}{|T_{\text{cal}}|} \sum_{t \in T_{\text{cal}}} \left( z_{i,t} - \sum_{j \in N_a} w_j z_{j,t} \right)^2 .$$

The intercept is represented by calibration centering in the residual definition of Section 2; no post-anchor outcome is supplied to the fit.

Fixed-weight MetaShift begins from the pre-event reliability preference

$$\pi_{ij} \propto \max(\rho_{ij}, 0)^2 \exp(-d_{ij}/50),$$

where $\rho_{ij}$ is pre-event log-scale correlation and $d_{ij}$ is distance in km. Its real-audit objective is

$$\min_{\substack{w_j \geq 0 \\ \sum_j w_j = 1}} \text{MSE}_{T_{\text{cal}}}(z_i, Z_N w) + 0.1 \sum_j w_j^2 + 0.1 \sum_j (w_j - \pi_{ij})^2.$$

In the benchmark, cross-validated MetaShift chooses from a frozen penalty grid using the final 45 pre-event days and refits on the supplied calibration period. It never sees post-anchor data during selection.

## 6.4 Two-level donor rule

The AQS protocol distinguishes event eligibility from date-level calculation feasibility (Table 6). This preserves the meaning of a cross-site comparison even when a small number of otherwise eligible donors are temporarily missing.

Screening and correlation assembly scale with the available pre-event target–donor pairs, with deterministic donor ranking adding $O(D \log D)$. The retained donor set is capped at three to five sites.

**Table 6:** Two-level donor rule in the AQS deployment audit.

| Stage | Requirement | Purpose |
|---|---|---|
| Event eligibility | At least three distinct physical donor sites pass the geographic, stability, paired-day, and correlation screens. | Avoids counting multiple POCs from one site as independent geographic evidence. |
| Date-level calculation | At least two eligible donor sites are observed, with at least 30 retained observations in every required window. | Ensures a numerical residual can be calculated after date-level availability filtering. |

**Table 7:** Predeclared primary evidence-tier rules and abstention outcomes.

| Required diagnostic or condition | Threshold | Audit outcome |
|---|---|---|
| Complete common-method comparison | required | Missing input ⇒ inconclusive |
| Selection-aware nested interval | required | Unavailable ⇒ inconclusive |
| Unique stable post-transition time placebos | 50 | Fewer dates ⇒ inconclusive |
| Raw time-placebo probability | 0.10 | Above cutoff ⇒ not supported |
| BH-adjusted time-placebo q value | 0.10 | Unavailable ⇒ inconclusive; above cutoff ⇒ not supported |
| Fixed-weight conditional diagnostic interval | excludes 0 | Otherwise ⇒ not supported |
| Leave-one-donor-out direction fraction | 0.90 | Below cutoff ⇒ not supported |
| All available required conditions | pass | Supported candidate discontinuity |

*Note.* The rules synthesize observational diagnostics; they are not a supervised classifier, instrument-fault label, or physical-causality test. The nested interval is required for tier assignment but is not claimed to have synthetic coverage calibration.

### 6.5 Intervals, falsification, and audit dispositions

For fixed donor weights, the primary interval resamples pre- and post-anchor residuals using circular moving blocks of seven days and 1,000 fixed-seed repetitions. It is conditional on previously chosen donors and weights. The selection-aware nested procedure re-block-resamples the calibration matrix, recomputes eligibility, selects donors from the fixed candidate pool, refits weights, and resamples the pre/post residual windows. Both are diagnostic intervals in the real audit, not calibrated estimates of physical bias.

The audit also computes stable post-transition time placebos, donor-as-treated placebos, a date-resampling comparison, and leave-one-donor-out refits. These are falsification and robustness diagnostics; they neither create random assignment nor identify a physical mechanism.

An event is inconclusive if a required comparative or diagnostic input is unavailable. It is not supported when all required diagnostics are available but at least one fails. A supported candidate satisfies every required diagnostic. Benjamini–Hochberg adjustment is only an exploratory multiple-screening quantity [12]; the resulting tiers are observational audit summaries rather than labels for instrument failure, replacement, bias, or pollution correction.

## 7 Experimental design

### 7.1 Primary target-fixed evaluation

The primary answerability experiment was implemented and protocol-checked before evaluation outputs were created. It uses bounded synthetic panels only; AQS data, AQS metadata, earlier evidence releases, and taxonomy decisions are not inputs. Each component has 300 dates, one target, four donors, a fixed anchor day, equal donor weights, and independently generated bounded common and idiosyncratic innovations.

Calibration contains 120 components and evaluation contains 360 disjoint components. The complete design has 76,800 calibration pairs and 230,400 evaluation pairs. Each component contributes 640 matched pairs, and each pair contains two scope arms:

**Sample accounting.** 120 calibration components $\rightarrow$ 76,800 pairs; 360 held-out components $\rightarrow$ 230,400 pairs $\rightarrow$ 460,800 scope arms. Uncertainty is resampled at the component level because pairs and arms within a component share its generated panel.

The $5 \times 4 \times (2^5) = 640$ grid crosses five nominal participation values $q \in \{0, .25, .50, .75, 1\}$ and four signal strengths $H \in \{.04, .08, .12, .20\}$ with five binary stress factors: donor mismatch, availability loss, one-donor contamination, a common nonlinear raw-scale field, and bounded noise. Every cell produces one local and one shared arm. The target arrays must be exactly identical within a pair; at $q = 0$, donor arrays must be identical as well. The protocol fails closed on target-identity violations, footprint leakage, stale hashes, incomplete cells, calibration/evaluation mixing, or outcome-dependent configuration changes.

### 7.2 Policies, thresholds, and uncertainty

A calibration-only threshold selects the comparative forced policy. Confidence cutoffs are selected from 101 calibration quantiles separately for each predeclared tolerance. Evaluation reports every policy at every reporting tolerance; it does not select a new policy or retune a cutoff after outcomes. The certificate threshold is analytic and uses the declared bounds plus realized retained-date availability and panel values required by the clipping-aware raw-field bound. The ambiguous partial-scope region consists of all $q > 0$ pairs with nonpositive structural margin; the certificate must abstain there. The $q = 0$ stratum is retained separately as observationally unanswerable.

The unit of uncertainty is the evaluation component rather than a scope arm, date, or grid row. A 1,000-repetition percentile component bootstrap with a prespecified seed reports policy risk–coverage, finite-policy envelopes, certificate behavior, and subgroups without collapsing them into one favorable score.

### 7.3 Independent stable-regime benchmark

The AQS-based stable-regime benchmark supplies a separate, known-effect comparison of cross-site and single-series methods. It contains 146 physical target sites: 66 for calibration and 80 for held-out evaluation. Whole connected components of the complete target-plus-donor physical-input graph are assigned to one split, so no physical input site crosses calibration and evaluation.

The benchmark injects target-only additive, proportional, gradual-drift, temporary-step, and variance changes, together with matched target-and-donor regional variants. Each variant has 400 held-out samples. Ranking thresholds are selected only on calibration cases. Local-effect MAE, AUPRC, macro-F1, precision, recall, and regional false-positive rate are reported as complementary metrics; the variance-only condition has no defined level-effect truth and therefore no MAE.

### 7.4 AQS deployment audit and interval protocol

The AQS audit applies the distinct-physical-donor construction, pre-event fitting boundaries, and effect windows to all 563 metadata anchors. Cross-validated MetaShift is benchmark-only and is not introduced into the real audit after its outcomes are known. Predeclared sensitivity analyses vary one factor at a time and record availability and interpretation dependence, not a search for a favorable setting.

Fixed-weight interval coverage is evaluated on held-out synthetic effects with fixed confidence level, seven-day blocks, perturbations, and 1,000 repetitions. The prespecified full nested protocol exceeded the fixed computational budget and was not executed; consequently, no selection-aware coverage claim is made. In the real audit, the calibration slice $t_0 - 180$ through $t_0 - 15$ overlaps the displayed pre-anchor slice $t_0 - 60$ through $t_0 - 1$ by 46 days. No post-anchor data enter the fit, but the displayed pre-anchor residuals are not an independent validation sample.

Same-site POC records, narrow hourly windows, QA availability, and targeted official-document review provide contextual checks with known limits. A separate parameter-88502 pipeline uses its own scan and audit; it is never pooled with the primary 88101 results.

# 8 Results

## 8.1 RQ1: Comparative information expands the finite-policy frontier only at a relaxed tolerance

**Answer.** On the target-fixed held-out grid, the target-only forced policy answered every scope arm but had 50% observed conditional scope error. Thus no predeclared target-only policy had positive finite-policy coverage at any reported tolerance below 0.5. The comparative forced policy also answered every arm but had 27.5% observed error. Its confidence-selective policies calibrated at .01, .05, and .10 answered 0 held-out arms. Only the policy calibrated at .20 reached a positive comparative envelope: 60.4% coverage at 19.5% observed conditional error.

**Table 8:** Frozen finite-policy scope-answerability envelope on the held-out target-fixed evaluation components. Parentheses give observed conditional scope error for the policy attaining the displayed coverage; – means no positive-coverage candidate qualified. The certificate-assisted channel uses synthetic design information and is not an operational deployment channel. Its 0 observed errors are counted among 179,994 answered scope arms, not estimated population risk.

| Tolerance $\alpha$ | Target-only | Comparative | Certificate-assisted |
|---|---|---|---|
| 1% | 0% | 0% (–) | 39.1% (0 observed errors) |
| 5% | 0% | 0% (–) | 39.1% (0 observed errors) |
| 10% | 0% | 0% (–) | 39.1% (0 observed errors) |
| 20% | 0% | 60.4% (19.5%) | 39.1% (0 observed errors) |

At $\alpha = .20$, the achieved finite-policy Scope Answerability Gain from the target-only to comparative channel is 60.4%. Its component-bootstrap interval is $[60.1\%, 60.7\%]$. This is a comparison among frozen policies on held-out synthetic components, not an estimate of a universal optimum or future population risk.

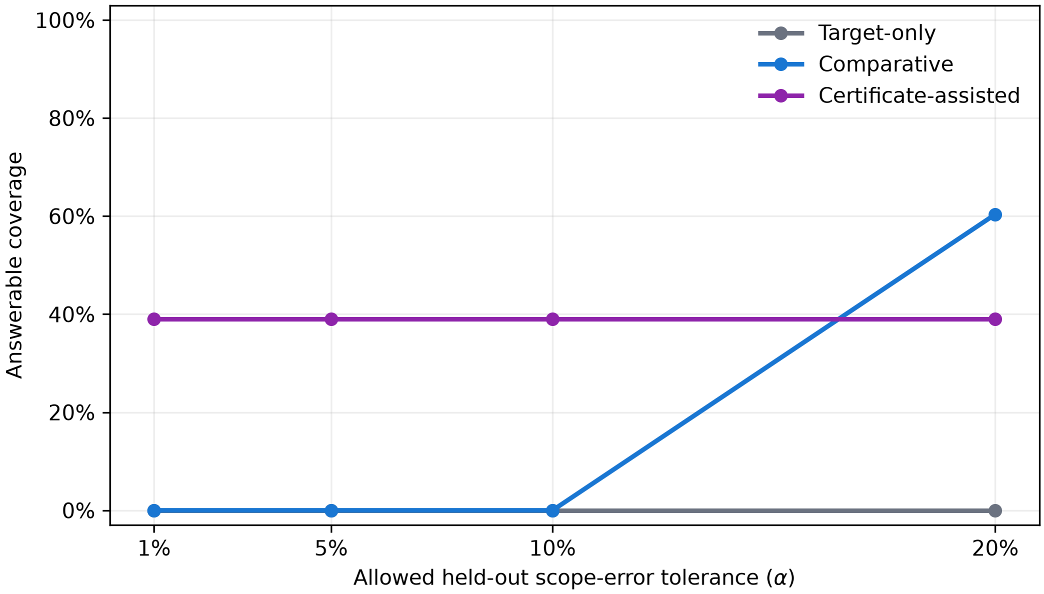


**Figure 4:** Held-out finite-policy answerability frontiers by information channel. Target-only policies do not reach positive coverage at the four reported tolerances. Comparative coverage emerges only at $\alpha = .20$. The certificate-assisted policy is plotted separately because it uses bounded synthetic-design information, not a real-network deployment channel.

## 8.2 RQ2: Structural separation maps a retained abstention boundary

**Answer.** The structural interval-separation certificate answered 89,997 of 230,400 matched pairs (39.1%), with 0 observed errors among 179,994 answered scope arms. It recovered 97.7% of pairs in the predeclared simulation-information reference region and abstained on the remaining 140,403 pairs (60.9%), including all 46,080 observationally indistinguishable $q = 0$ negative controls. It had 0 observed envelope violations. These finite-grid observations hold only for the bounded generator: they neither establish zero population risk nor yield a deployable real-network certificate.

**Table 9:** Certificate behavior by nominal donor participation on held-out target-fixed pairs. Reference-region recovery is the share of the predeclared simulation-information reference region recovered by the certificate; it is undefined at $q = 0$ because that negative-control stratum has no reference-region pairs. Across these strata, 0 observed errors means 0 of 179,994 answered scope arms, not population risk.

| Participation | Pair coverage | Observed error | Reference recovery |
|---|---|---|---|
| $q = 0\%$ | 0% | – | – |
| $q = 25\%$ | 10.9% | 0 observed errors | 87.5% |
| $q = 50\%$ | 38.3% | 0 observed errors | 94.2% |
| $q = 75\%$ | 66.4% | 0 observed errors | 98.8% |
| $q = 100\%$ | 79.7% | 0 observed errors | 100% |

The certificate coverage increases with nominal donor participation and signal, but it is intentionally incomplete under contamination, nonlinear raw fields, and heteroskedastic noise. In particular, the $q = 0$ stratum contains 46,080 matched pairs with identical target and donor observations in both arms, produces 0 certificate answers, and has no reference-region pair authorized for an answer. The result does not declare all real partial scope objectively ambiguous; it marks where this sufficient bound does not authorize an answer.

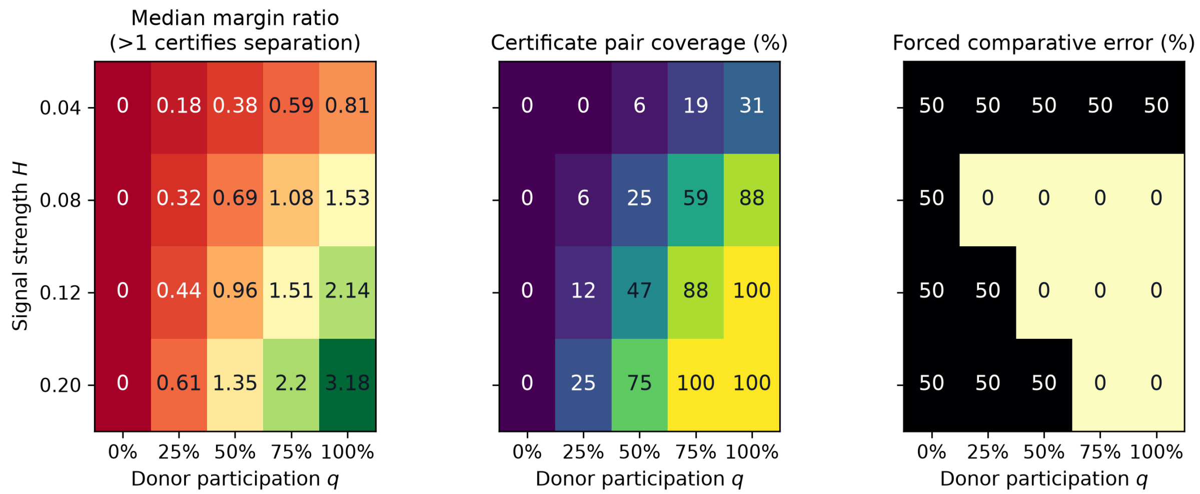


**Figure 5:** Structural answerability boundary on the held-out grid. From left to right, the panels show the median normalized structural margin, certificate answered-pair coverage, and forced-comparative scope error. Values above one in the first panel meet the sufficient interval-separation condition. Each cell averages over the five adverse comparison factors; $q = 0$ remains the observational-identity negative control.

**Table 10:** Predeclared failure and abstention accounting on 230,400 held-out target-fixed matched pairs. These rows are retained rather than converted into a favorable single score. The rows overlap and must not be summed: the $q = 0$ row is a subset of the nonpositive-margin row, while arm-level error rows use a different denominator from pair-level rows.

| Outcome | Count | Interpretation |
|---|---|---|
| Target and donor observations identical at $q = 0$ | 46,080 | negative-control abstention stratum |
| Nonpositive structural margin | 140,403 | certificate must abstain |
| Certificate-answered pairs | 89,997 | 0 errors among 179,994 answered arms |
| Envelope violations | 0 | none observed |
| Comparative-forced errors | 126,764 | among 460,800 forced scope arms |
| Confidence policy at calibration $\alpha = 0.05$ | 0 | no answered arms |

### 8.3 RQ3: Cross-site comparison separates the constructed task, but estimator superiority is not supported

**Answer.** In the independent stable-regime benchmark, all four cross-site methods have aggregate AUPRC between 0.884 and 0.915, whereas the five target-only methods range only from 0.500 to 0.501 (Table 16 in Appendix A.1). This positive information contrast is the intended behavior of the constructed local-versus-regional task, not a generic change-detection claim.

**Table 11:** Independent stable-regime synthetic evaluation.

| Method | MAE | AUPRC | Macro-F1 | Regional FPR |
|---|---|---|---|---|
| Standard SC | 0.100 | 0.915 | 0.816 | 0.140 |
| MetaShift fixed | 0.103 | 0.898 | 0.795 | 0.150 |
| MetaShift CV | 0.097 | 0.902 | 0.809 | 0.135 |
| Nearest-neighbor DiD | 0.111 | 0.884 | 0.789 | 0.075 |

*Note.* Metrics are aggregates over the frozen `stable_full_v2` evaluation set. Lower MAE and regional false-positive rate are preferable; higher AUPRC and macro-F1 are preferable.

Standard synthetic control has the strongest aggregate AUPRC and macro-F1. Cross-validated

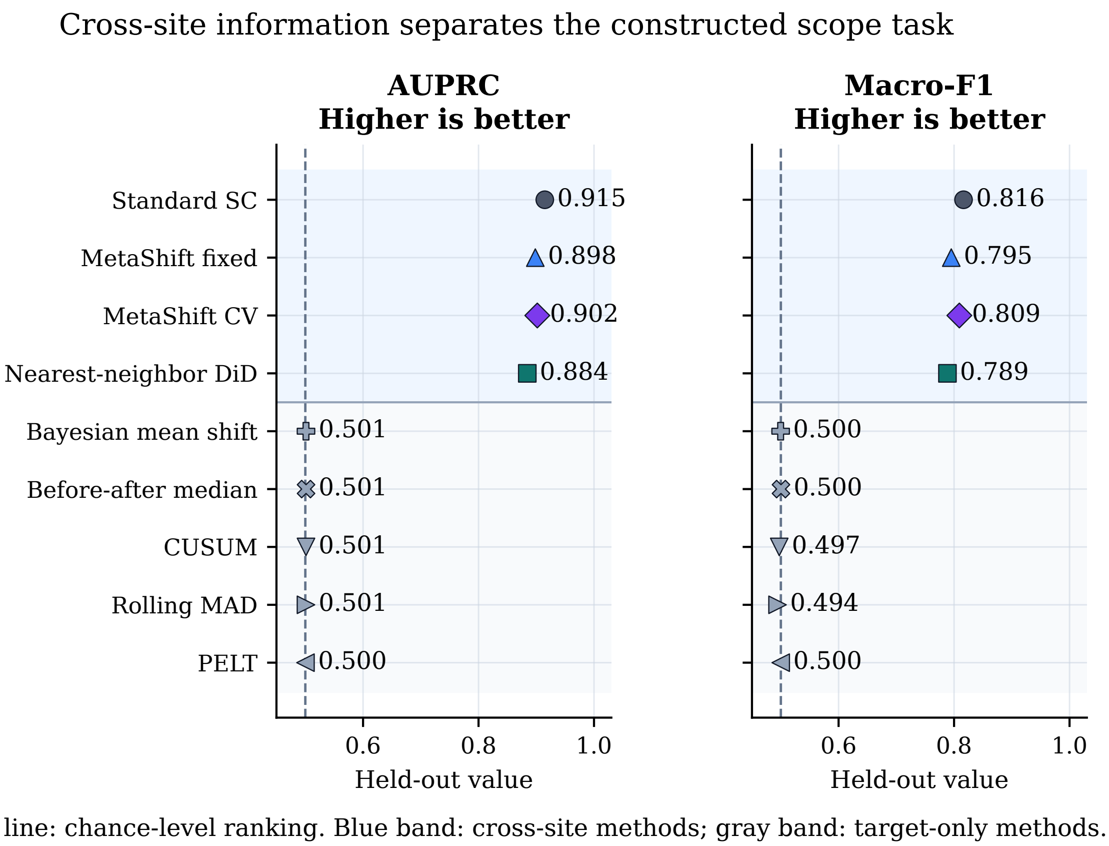


**Figure 6:** Cross-site versus target-only ranking on the independent held-out stable-regime benchmark. All four cross-site methods separate the constructed local-versus-shared scope task, while the five target-only methods remain near chance. The comparison demonstrates an information contrast in this benchmark, not a universal method ranking or a real-network mechanism result.

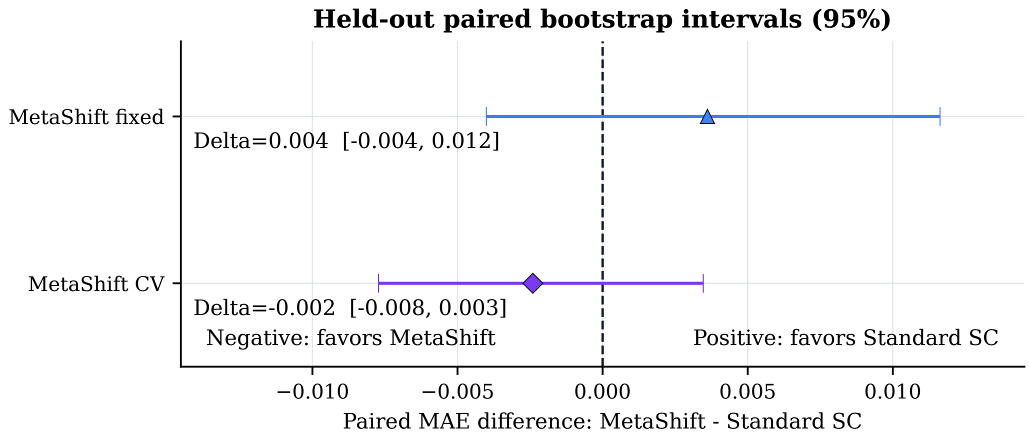


**Figure 7:** Paired event-cluster bootstrap intervals for MetaShift minus standard synthetic-control local-effect MAE. Negative values favor MetaShift; both intervals cross zero. The frozen evidence therefore does not support an aggregate MetaShift-superiority claim.

MetaShift has the smallest aggregate local-effect MAE, while nearest-neighbor difference-in-differences has the lowest regional false-positive rate. No method wins every metric. Most importantly, neither MetaShift variant has a confidence-supported aggregate MAE advantage over standard synthetic control: both paired component-bootstrap intervals include zero (Table 12).

**Table 12:** Paired event-cluster bootstrap for local-effect MAE differences.

| Comparison | Difference | 95% bootstrap interval | Clusters |
|---|---|---|---|
| MetaShift fixed minus Standard SC | 0.004 | [-0.004, 0.012] | 80 |
| MetaShift CV minus Standard SC | -0.002 | [-0.008, 0.003] | 80 |

*Note.* Difference is MetaShift minus standard synthetic control; negative values favor MetaShift on MAE. Both bootstrap intervals include zero.

### 8.4 RQ4: A qualified AQS comparison is available for 228 of 563 anchors

**Answer.** The distinct-physical-donor rule permits a complete common cross-site comparison for 228 of 563 anchors (40.5%). The other 325 donor-insufficient and 10 input-window failures are retained as abstentions, not dropped observations. This availability result is the AQS audit's primary deployment finding.

The exploratory evidence synthesis assigns 34 supported candidates, 122 not-supported events, and 407 inconclusive events. "Supported" means that all predeclared diagnostic conditions were available and passed; it does not confirm a physical failure, replacement, measurement bias, or pollution correction.

**Table 13:** Complete 88101 event audit and real-event diagnostics.

| Audit disposition | Anchors |
|---|---|
| Complete common-method comparison | 228 |
| Fewer than three distinct physical donors | 325 |
| Estimator input-window failure | 10 |
| Total | 563 |

| Method | Events | Fixed conditional interval excludes 0 | Median log effect |
|---|---|---|---|
| MetaShift fixed | 228 | 159 | -0.087 |
| Standard SC | 228 | 145 | -0.071 |
| Nearest-neighbor DiD | 228 | 134 | -0.075 |

*Note.* Nested selection-aware intervals are available for 227/228 complete events; 153 exclude zero. Their mean width is 0.191 log units versus 0.179 for fixed-weight MetaShift intervals. Leave-one-donor-out refits complete for 227 events, with 202 retaining direction under every donor removal. All intervals are diagnostic, not calibrated physical-bias confidence intervals.

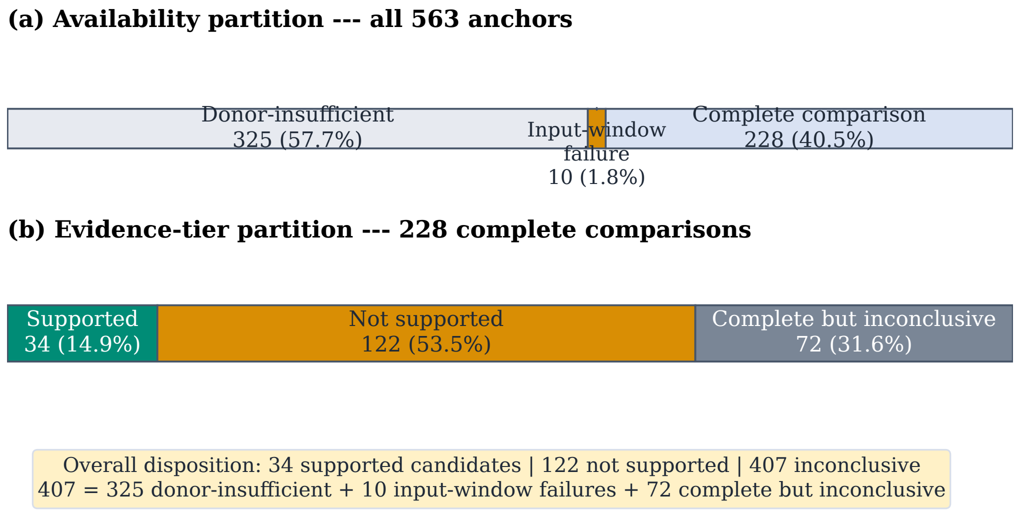


**Figure 8:** AQS audit accounting. Panel (a) partitions all 563 metadata anchors into donor-insufficient, input-window-failure, and complete comparison dispositions. Panel (b) partitions complete comparisons into protocol evidence tiers. These are audit dispositions, not diagnoses of physical monitor changes.

### 8.5 RQ5: Calibration and contextual checks limit, rather than validate, real-data interpretation

**Answer.** The fixed-protocol interval constructions do not support a generic calibration claim. On the held-out synthetic effects, nominal 95% fixed-weight coverage ranges from 63.9% to 67.3% over 3,200 instances per method, while nominal 90% split-conformal coverage ranges from 98.8% to 99.6% and is conservative.

Of 11 same-site POC candidates, 9 have usable paired hourly windows and 8 agree in daily/hourly direction. Targeted official-document review finds 0/20 dated site-specific confirmations. Missing documentation and unavailable POC data remain unavailable evidence; they are not evidence that nothing changed.

The real-data date-resampling diagnostic holds each event's pre-transition donor weights fixed and compares the mean standardized score at the metadata anchor with one randomly selected stable post-transition date per eligible event. Under the within-event exchangeability diagnostic null, its statistic is the difference between the anchor mean and sampled-placebo mean; the reported one-sided tail probability is $(1 + \#\{\text{sampled placebo mean} \geq \text{anchor mean}\})/201 = 0.00498$. It is a comparison with stable dates, not a causal probability or a physical-mechanism test.

**Table 14:** Placebo, external-document, and same-site POC evidence boundaries.

| Diagnostic | Value |
|---|---|
| Complete time-placebo events with at least 50 dates | 157 |
| Complete time-placebo events with 100 dates | 128 |
| Raw time-placebo probabilities at most 0.10 | 71 |
| BH q values at most 0.10 | 40 |
| Donor-as-treated placebo records | 866 |
| Date-resampling permutations | 200 |
| Dated site-specific external confirmations | 0/20 |
| Usable hourly same-site POC comparisons | 9/11 |
| Daily/hourly POC direction agreements | 8/9 |
| Standard SC 90th-percentile risk-gate retention | 75/80 |

*Note.* Time and donor placebos are diagnostic falsifications. Same-site POC and document review provide contextual evidence only, not instrument ground truth.

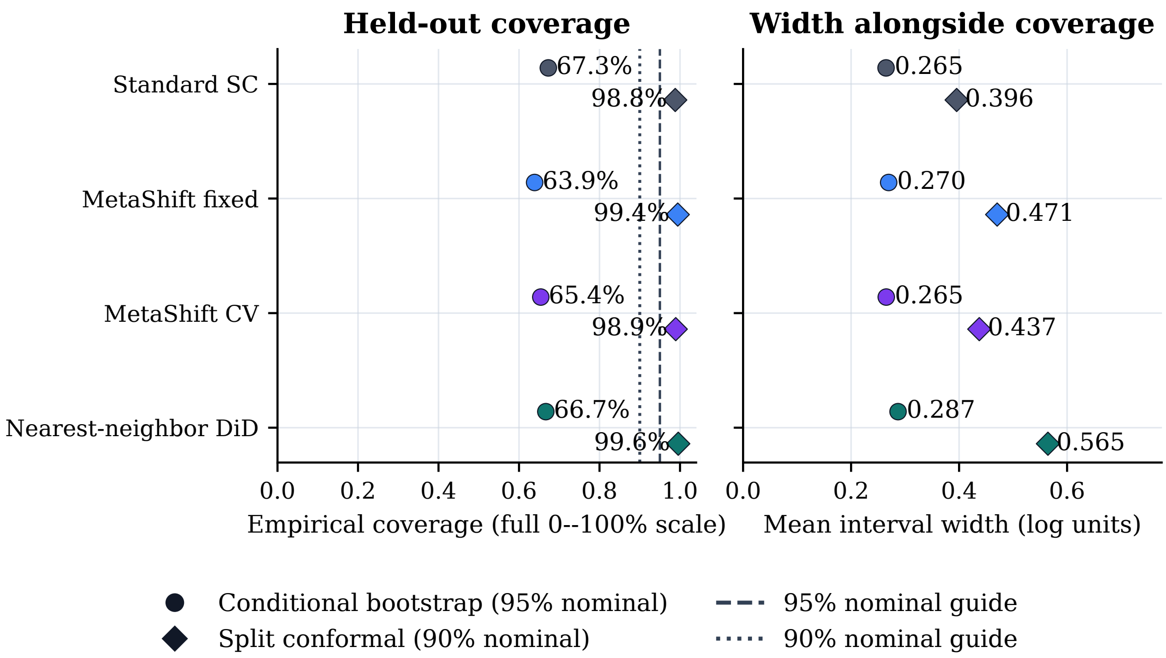


**Figure 9:** Held-out synthetic interval coverage and mean width. Fixed conditional intervals under-cover their nominal 95% target, whereas split-conformal intervals over-cover their nominal 90% target. The figure reports a calibration limitation, not a real-event confidence guarantee.

# 9 Discussion

## 9.1 Main answer

The central result is conditional rather than triumphant: a target-only detector can recognize a change, yet cannot resolve its local-versus-shared scope when matched target observations carry no scope information. Comparative information enlarges the achieved held-out answerability frontier only at the least strict reported tolerance. The experiment therefore rejects both extremes: target information alone is insufficient in the stipulated paired worlds, and donor comparison does not automatically make every scope question answerable.

## 9.2 Why comparison sometimes fails

The partial-scope grid makes the failure boundary visible. Low donor participation, weak target signal, missing donors, contamination, nonlinear raw-scale leakage, and noise shrink the interval gap on which the certificate relies. A nonpositive structural margin is not treated as a low-confidence label: the certificate abstains. This is useful precisely because the failed and unanswerable cells remain in the result rather than being discarded to improve a classifier score.

## 9.3 An evaluation-design lesson

Holding the target path fixed makes the information question testable. A target-only rule cannot receive credit for exploiting an accidental target-side cue, while donor-side variation can be made explicit and degraded continuously. This design does not make the generator realistic by itself; it makes the boundary of one carefully stated scope question auditable. The negative controls, intermediate participation levels, and retained failures are therefore part of the result rather than implementation details.

### 9.4 Why estimator complexity did not win

The stable-regime benchmark confirms that cross-site information is useful for the constructed scope task, but it does not support a general MetaShift advantage. Standard synthetic control leads aggregate ranking and macro-F1; cross-validated MetaShift has the best point estimate for local-effect MAE, but its paired uncertainty interval versus standard synthetic control crosses zero. This is plausible within the stipulated benchmark: once qualified donor paths are available, a simple donor-weighted control already captures much of the cross-site contrast, and an additional reliability prior can trade bias against variance differently across perturbation families. This explanation is not a general ranking claim; the frozen comparison leaves the absence of a confidence-supported MetaShift advantage intact. The appropriate contribution is consequently the information-and-abstention framework, not a claim that a more complex estimator is universally better.

### 9.5 Implications for real audits

The AQS inventory measures whether a qualified comparison can be formed, not whether a monitor physically changed. Its many donor-insufficient and input-window-failure dispositions quantify where this audit cannot answer. A supported candidate only prioritizes an anchor for technical-record review. Station histories, maintenance records, and human domain review remain necessary for a mechanism claim; neither metadata anchors, residuals, placebos, nor same-site POC agreement substitute for them.

## 10 Limitations

### 10.1 Scope of the answerability claims

Table 15 separates the assumptions supporting the target-fixed answerability statements from claims they do not support. The certificate relies on declared bounded generator quantities, including participation, mismatch, leakage, and noise envelopes. It is therefore a synthetic-design-assisted sufficient condition, not a confidence guarantee, an empirical $O_E$ evidence channel, or a deployable sensor-network rule.

### 10.2 Limits of the AQS audit

AQS Method Code transitions are metadata anchors, not physical ground truth. Pre-event matching, distinct physical donors, time and donor placebos, and leave-one-donor-out checks constrain obvious comparison failures, but do not identify a unique cause of a residual. The $t_0 - 180$ through $t_0 - 15$ calibration slice overlaps the displayed $t_0 - 60$ through $t_0 - 1$ pre-anchor slice by 46 days. No post-anchor outcome enters fitting, but the displayed pre-anchor residuals are consequently not an independent validation sample.

Fixed-weight moving-block intervals under-cover known synthetic effects, and split-conformal intervals over-cover them. The unexecuted full selection-aware synthetic coverage study was outside the fixed computational budget. Thus real-event intervals remain diagnostic descriptions, not coverage-calibrated confidence intervals for physical measurement bias.

**Table 15:** Assumptions and claim boundaries for the target-fixed scope-answerability experiment.

| Assumption | Supports | Does not support |
|---|---|---|
| Balanced labels, matched target laws, and no target-side leakage | Target-only impossibility in the stipulated paired worlds. | Target-only impossibility for arbitrary priors, leaked features, or AQS records. |
| Nested target-only and comparative channels under one law | Policy-set monotonicity under information refinement. | Guaranteed improvement by a fitted classifier. |
| Analysis-scale intervention, fixed weights, and arm-invariant retained dates | The exact construction-specific residual identity. | Exact raw-scale algebra after nonlinear transformation or clipping. |
| Declared bounded mismatch, noise, leakage, and contamination envelopes | Interval separation when $\kappa > 0$. | Unbounded-noise, population-risk, or real-network guarantees. |
| Known synthetic participation and bounds | Certificate coverage and reference-region recovery. | An operational external-evidence policy. |

### 10.3 Transfer and human-review requirements

The archive years, $PM_{2.5}$ parameter, U.S. network, availability rules, donor radius, and synthetic intervention families constrain transfer. The secondary 88502 pipeline demonstrates only sparse separate-pipeline feasibility. A physical mechanism, equipment diagnosis, or causal measurement-bias claim would require dated station histories, calibration and maintenance records, independently reviewed taxonomy, and human domain judgment. The complete threats-to-validity table and applicability map are retained in the appendix rather than omitted. No taxonomy-stratified analysis is reported.

## 11 Conclusion

MetaShift-Bench studies a question that ordinary change detection leaves open: when can an observed change be called local rather than shared? Its target-fixed paired design turns that question into a selective prediction problem. At a stated tolerated scope error, the answerability frontier measures how much of the stipulated task a channel can answer; the structural certificate abstains where bounded comparison evidence cannot separate the two scopes.

The held-out findings are deliberately mixed. Target-only scope inference has no positive finite-policy coverage below one-half error in the matched paired construction. Comparative information gains coverage only at the least strict reported tolerance. The certificate has no observed errors where its declared intervals separate, but abstains on most pairs and is not a real-network deployment policy. The separate AQS audit records comparison availability and diagnostic abstentions, while preserving the negative result that MetaShift does not have a confidence-supported aggregate advantage over standard synthetic control.

These results support a bounded principle: an audit should state what information its scope decision requires and should abstain when that information is insufficient. They do not establish real monitor mechanisms, population-risk guarantees, physical measurement bias, or corrected concentrations. Those questions require independently documented records and human review. Reliable audits should not merely detect a change; they should also declare when the available evidence can, and cannot, answer where that change belongs.

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# A Supplementary evidence and protocols

## A.1 Complete method comparisons

Table 16 restores all nine frozen stable-regime benchmark methods. The five target-only methods remain near chance on the constructed local-versus-regional ranking task; this does not mean they are unusable for ordinary single-series change detection. N/A local-effect MAE means that a method does not estimate the level-effect target for that perturbation design.

**Table 16:** Complete frozen aggregate comparison of all benchmark methods.

| Method | Local-effect MAE | AUPRC | Macro-F1 | Regional FPR |
|---|---:|---:|---:|---:|
| Standard SC | 0.09983 | 0.91476 | 0.81641 | 0.140 |
| MetaShift fixed | 0.10344 | 0.89832 | 0.79515 | 0.150 |
| MetaShift CV | 0.09742 | 0.90178 | 0.80916 | 0.135 |
| Nearest-neighbor DiD | 0.11150 | 0.88443 | 0.78852 | 0.075 |
| Bayesian mean shift | 0.24672 | 0.50116 | 0.49962 | 0.542 |
| Before-after median | 0.25330 | 0.50125 | 0.50024 | 0.505 |
| CUSUM | N/A | 0.50117 | 0.49720 | 0.581 |
| Rolling MAD | N/A | 0.50112 | 0.49427 | 0.402 |
| PELT | N/A | 0.50025 | 0.49976 | 0.522 |

*Note.* All values are aggregates over the held-out stable-regime evaluation partition. Lower MAE and regional FPR are preferable; higher AUPRC and macro-F1 are preferable. N/A means that the method supplies no local-effect magnitude estimate for that perturbation design, not that the method was omitted.

For legibility, the complete perturbation matrix groups rows by family on ordinary portrait pages; no method, perturbation family, or unfavorable cell is removed.

**Table 17:** Perturbation-family-specific held-out synthetic metrics for all methods.

| Method | MAE | AUPRC | Macro-F1 | Regional FPR |
|---|---:|---:|---:|---:|
| **Additive step** | | | | |
| Standard SC | 0.07788 | 0.99946 | 0.93092 | 0.138 |
| MetaShift fixed | 0.08373 | 0.99695 | 0.87969 | 0.233 |
| MetaShift CV | 0.07680 | 0.99801 | 0.90161 | 0.193 |
| Nearest-neighbor DiD | 0.08680 | 0.99210 | 0.93614 | 0.105 |
| Bayesian mean shift | 0.22860 | 0.50497 | 0.41031 | 0.890 |
| Before-after median | 0.22774 | 0.50497 | 0.44778 | 0.808 |
| CUSUM | N/A | 0.50497 | 0.43785 | 0.833 |
| Rolling MAD | N/A | 0.50497 | 0.49862 | 0.553 |
| PELT | N/A | 0.50497 | 0.42673 | 0.858 |
| **Proportional step** | | | | |
| Standard SC | 0.07635 | 0.85856 | 0.76319 | 0.188 |
| MetaShift fixed | 0.08016 | 0.84617 | 0.78293 | 0.090 |
| MetaShift CV | 0.07457 | 0.84296 | 0.78357 | 0.103 |
| Nearest-neighbor DiD | 0.08287 | 0.82011 | 0.70938 | 0.065 |
| Bayesian mean shift | 0.23444 | 0.50497 | 0.48849 | 0.350 |
| Before-after median | 0.22774 | 0.50497 | 0.46944 | 0.260 |
| CUSUM | N/A | 0.50497 | 0.48467 | 0.328 |
| Rolling MAD | N/A | 0.50497 | 0.40894 | 0.108 |
| PELT | N/A | 0.50497 | 0.48326 | 0.320 |
| **Gradual drift** | | | | |

*Note.* Each family aggregates its target-only local perturbation and its matched target-and-donor regional variant. Consequently, regional FPR is assessed within the same paired family; the rows are not local-only effect estimates. N/A has the same meaning as in Table 16.

*Table 17, continued*

| Method | MAE | AUPRC | Macro-F1 | Regional FPR |
|---|---|---|---|---|
| Standard SC | 0.08321 | 0.99710 | 0.93985 | 0.110 |
| MetaShift fixed | 0.08551 | 0.99224 | 0.89419 | 0.193 |
| MetaShift CV | 0.07958 | 0.99521 | 0.92082 | 0.153 |
| Nearest-neighbor DiD | 0.08746 | 0.98570 | 0.93248 | 0.085 |
| Bayesian mean shift | 0.24325 | 0.50497 | 0.45055 | 0.800 |
| Before-after median | 0.23075 | 0.50497 | 0.47143 | 0.733 |
| CUSUM | N/A | 0.50497 | 0.45322 | 0.793 |
| Rolling MAD | N/A | 0.50497 | 0.49969 | 0.475 |
| PELT | N/A | 0.50497 | 0.49992 | 0.513 |
| **Temporary step** | | | | |
| Standard SC | 0.16187 | 0.98113 | 0.91246 | 0.108 |
| MetaShift fixed | 0.16436 | 0.96551 | 0.88871 | 0.130 |
| MetaShift CV | 0.15875 | 0.96994 | 0.90372 | 0.113 |
| Nearest-neighbor DiD | 0.18885 | 0.93733 | 0.84114 | 0.065 |
| Bayesian mean shift | 0.28058 | 0.50497 | 0.49930 | 0.463 |
| Before-after median | 0.32696 | 0.50497 | 0.49614 | 0.588 |
| CUSUM | N/A | 0.50497 | 0.48686 | 0.660 |
| Rolling MAD | N/A | 0.50497 | 0.45747 | 0.780 |
| PELT | N/A | 0.50497 | 0.47750 | 0.708 |
| **Variance increase** | | | | |
| Standard SC | N/A | 0.58680 | 0.48825 | 0.158 |
| MetaShift fixed | N/A | 0.57484 | 0.45372 | 0.105 |
| MetaShift CV | N/A | 0.57314 | 0.46902 | 0.113 |
| Nearest-neighbor DiD | N/A | 0.56157 | 0.42775 | 0.055 |
| Bayesian mean shift | N/A | 0.50056 | 0.45595 | 0.205 |
| Before-after median | N/A | 0.50741 | 0.42520 | 0.135 |
| CUSUM | N/A | 0.50470 | 0.48012 | 0.290 |
| Rolling MAD | N/A | 0.50191 | 0.39371 | 0.093 |
| PELT | N/A | 0.49172 | 0.45495 | 0.213 |

*Note.* Each family aggregates its target-only local perturbation and its matched target-and-donor regional variant. Consequently, regional FPR is assessed within the same paired family; the rows are not local-only effect estimates. N/A has the same meaning as in Table 16.

The reliability-prior ablations share the standard-control rows exactly to the configured numerical tolerance. They do not establish a stable MetaShift advantage, but make the frozen design variants available for inspection.

## A.2 Historical endpoint and supplementary answerability diagnostics

The immutable v0.4 construction is retained only as an endpoint sanity check. When its local and regional worlds hold the target sequence fixed, target-only classification has observed error 0.5, and its constructed comparative rule has zero observed error on that finite endpoint evaluation. It does not measure a risk–coverage frontier, calibration-selected abstention, partial scope, a failure boundary, or real measurement mechanisms.

Figure 13 gives the full frozen certificate-validity summary. Its reader-facing zero-error labels are observed finite-grid values, not population-risk estimates.

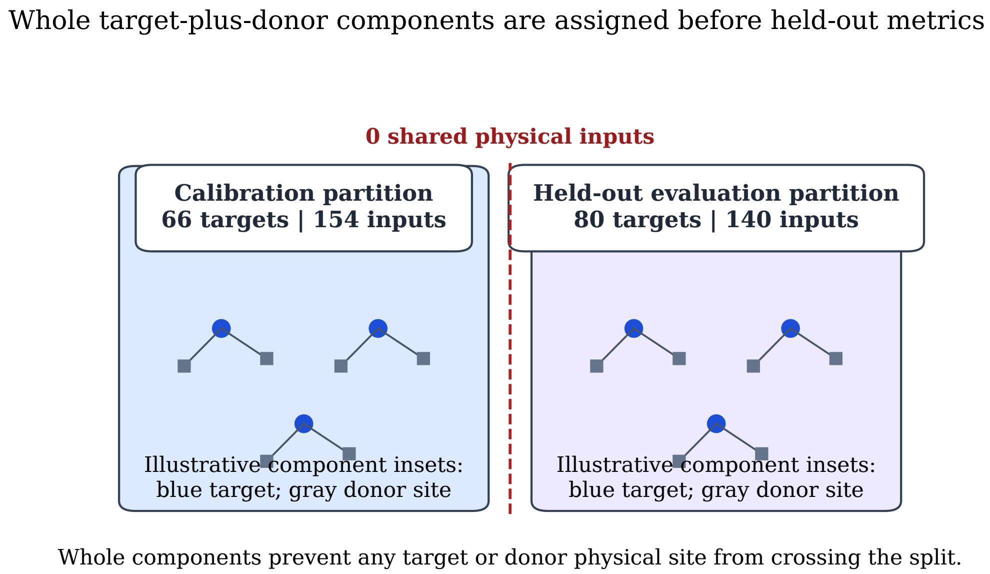


**Figure 10:** Connected-component split integrity for the stable-regime benchmark. No physical input site is shared between calibration and held-out evaluation.

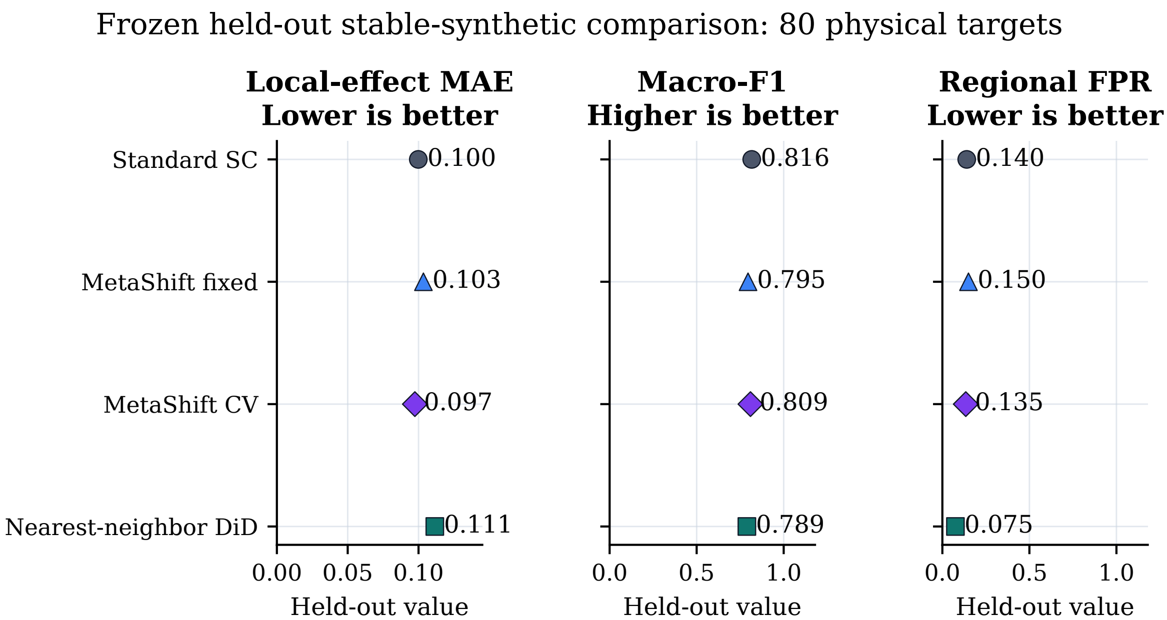


**Figure 11:** Held-out stable-regime comparison for the four cross-site methods. Dots show local-effect MAE, macro-F1, and regional false-positive rate. These panels do not form a composite ranking.

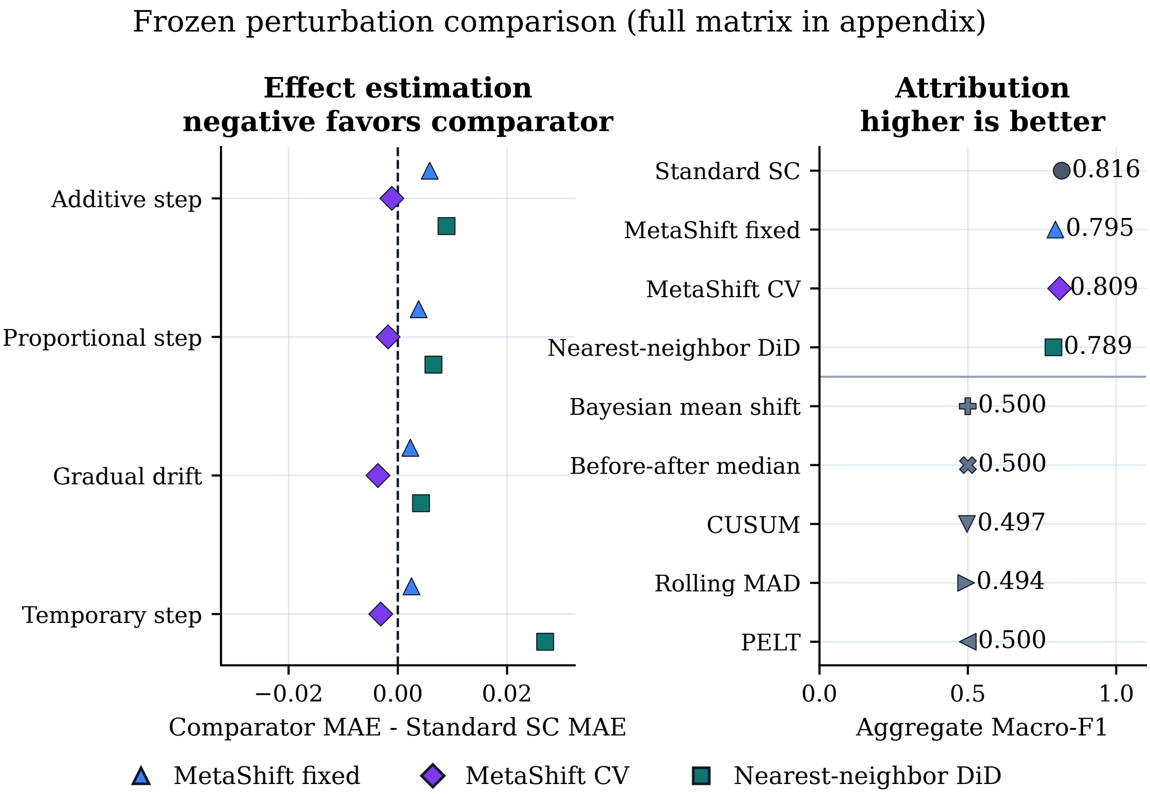


**Figure 12:** Perturbation-family comparison. The left panel gives local-effect MAE differences from standard synthetic control for cross-site comparators; the right panel gives aggregate macro-F1 for all nine methods. Variance-only local-effect MAE is undefined by design.

**Table 18:** Reliability-prior and regularization ablations on shared synthetic inputs.

| Variant | MAE | Macro-F1 | Regional FPR |
|---|---|---|---|
| Standard SC | 0.09983 | 0.81641 | 0.140 |
| No ridge penalty | 0.10289 | 0.80127 | 0.138 |
| Ridge = 0.01 | 0.10304 | 0.80263 | 0.130 |
| Add coverage | 0.10332 | 0.79862 | 0.135 |
| MetaShift full prior | 0.10344 | 0.79515 | 0.150 |
| No reliability-prior penalty | 0.10350 | 0.80167 | 0.133 |
| No correlation term | 0.10362 | 0.79858 | 0.133 |
| No distance term | 0.10370 | 0.80588 | 0.088 |
| Uniform prior | 0.10378 | 0.80521 | 0.112 |
| Ridge = 1.0 | 0.10500 | 0.80285 | 0.097 |
| Direct reliability | 0.10543 | 0.80574 | 0.105 |

*Note.* The common standard-synthetic-control records align exactly across the main and ablation experiments to tolerance $10^{-10}$.

Certificate behavior on held-out pairs

| Group | Pair coverage | Observed scope error | Envelope violations | Reference recovery |
|---|---|---|---|---|
| Overall | 39.1% | 0 observed errors | 0 observed errors | 97.7% |
| $q = 0\%$ | 0% | -- | 0 observed errors | -- |
| $q = 25\%$ | 10.9% | 0 observed errors | 0 observed errors | 87.5% |
| $q = 50\%$ | 38.3% | 0 observed errors | 0 observed errors | 94.2% |
| $q = 75\%$ | 66.4% | 0 observed errors | 0 observed errors | 98.8% |
| $q = 100\%$ | 79.7% | 0 observed errors | 0 observed errors | 100% |

**Figure 13:** Supplementary certificate-validity summary. "0 observed errors" denotes a finite-grid observation under the bounded generator, not a population-risk estimate.

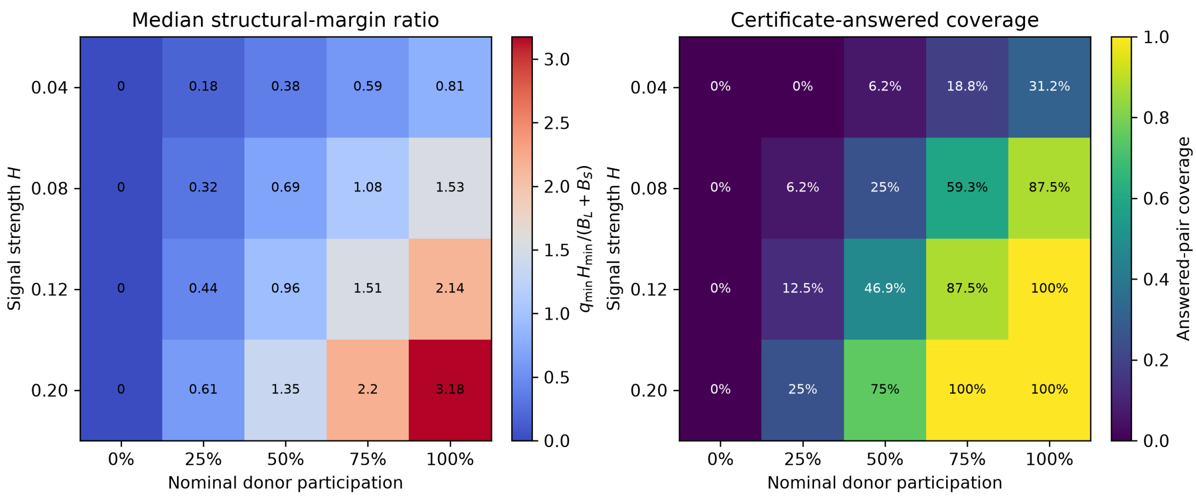


**Figure 14:** Supplementary normalized structural-margin phase diagram. The left panel shows median $q_{\min} H_{\min}/(B_L + B_S)$; values above one meet the sufficient interval-separation condition. The right panel shows certificate answered-pair coverage. Both panels average over the five adverse comparison factors and describe the bounded synthetic construction only.

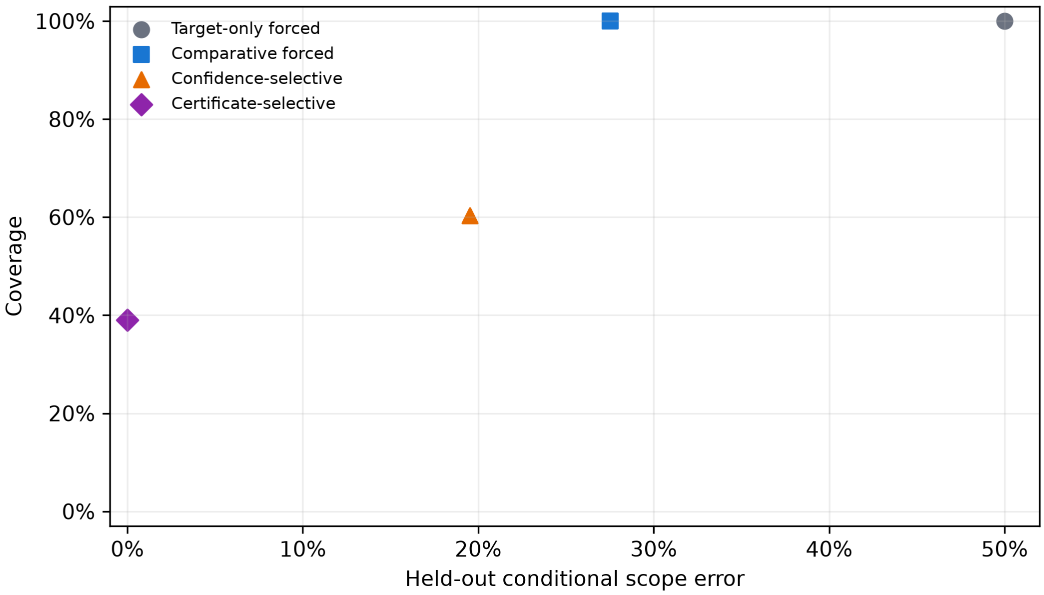


**Figure 15:** Supplementary risk–coverage positions of the predeclared held-out policies. Forced comparative classification has full coverage but high error; strict confidence policies abstain completely. The certificate has lower coverage than forced comparison but no observed errors under its bounded construction.

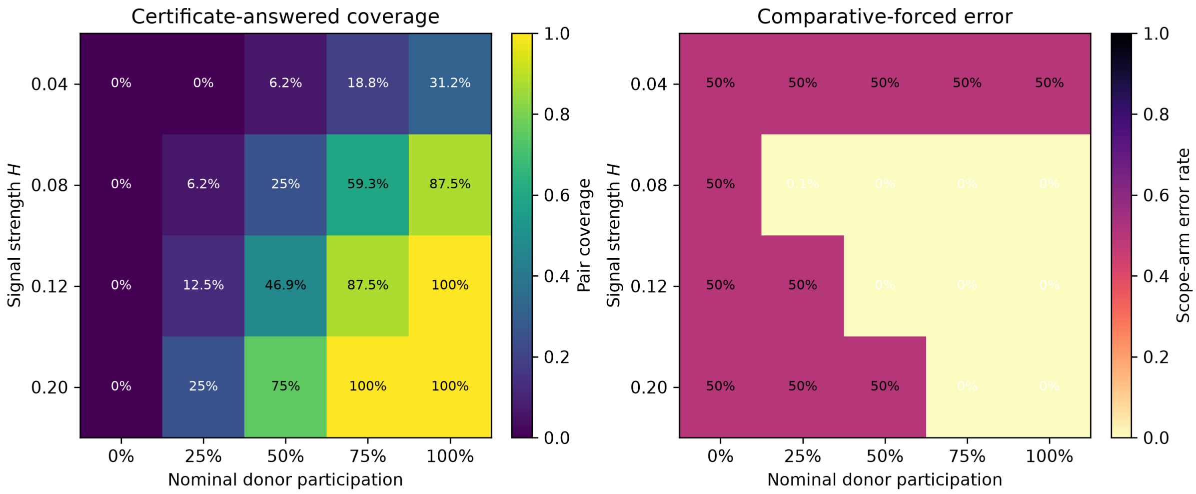


**Figure 16:** Supplementary retained failure map across signal and nominal donor participation. The left panel gives certificate coverage; the right panel gives comparative-forced error. Low signal and adverse comparison conditions remain in the evaluation rather than being filtered away. The $q = 0$ column is an observational-identity negative control.

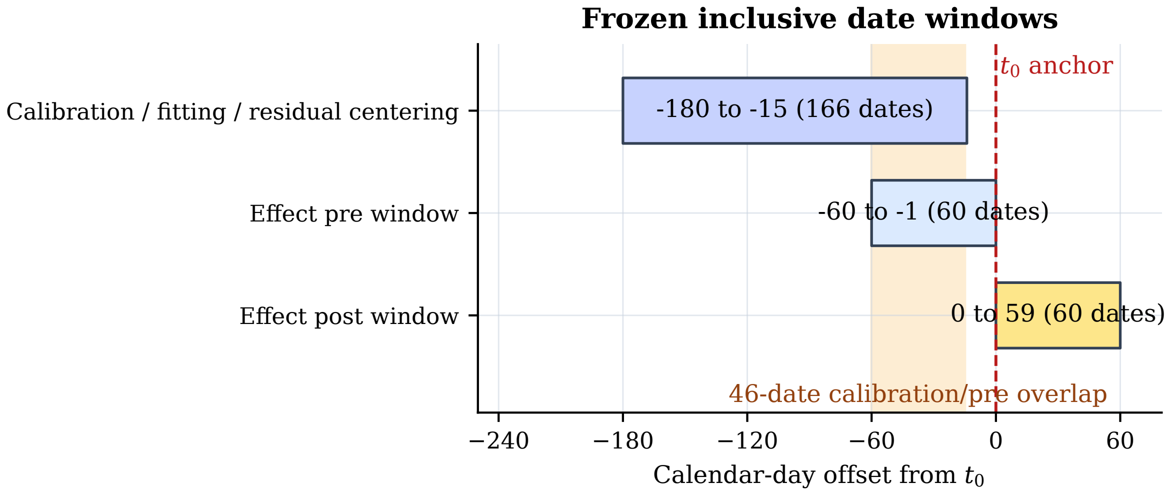


**Figure 17:** Calibration and displayed pre-anchor windows in the AQS audit. They overlap by 46 calendar dates, so displayed pre-anchor residuals are not an independent validation set. No post-anchor outcome enters fitting.

## A.3 AQS audit protocol

For each primary-parameter anchor $a = (s, p, t_0, m^-, m^+)$, the audit:

1. validates canonical daily observations and identifies a persistent reported Method Code transition;
2. constructs a geographic candidate pool at distinct physical sites, retaining at most one POC per site by the pre-event ranking rule;
3. records donor insufficiency if fewer than three qualified physical donors exist;
4. fits cross-site weights and the residual center only on the inclusive $t_0 - 180$ through $t_0 - 15$ calibration slice;
5. requires at least two available donors and 30 retained observations in each required residual window;
6. records an input-window failure if those calculation conditions fail;
7. computes diagnostic intervals, placebos, donor sensitivity, and protocol evidence tiers for complete cases; and
8. preserves complete cases, failures, and unavailable diagnostics.

This is an implementation specification for a metadata-anchored audit; it does not identify a physical mechanism.

**Deployment residual.** For a qualified target $i = (s, p)$ and donors $j \in N_a$, the AQS audit uses

$$r_{i,t} = z_{i,t} - \sum_{j \in N_a} w_{ij} z_{j,t} - \text{median}_{u \in T_{\text{cal}}} \left( z_{i,u} - \sum_{j \in N_a} w_{ij} z_{j,u} \right).$$

Its displayed effect is the post-minus-pre median residual on the log scale. It is a diagnostic local residual discontinuity, not a causal physical-bias estimate or a corrected concentration.

**Table 19:** Predeclared effect-window and reporting-scale sensitivity.

| MetaShift window | Complete events | Median log effect | Sign agreement with 60 days |
|---|---|---|---|
| 45 days | 224 | -0.08769 | 93.3% |
| 60 days | 228 | -0.08704 | reference |
| 90 days | 196 | -0.08064 | 93.9% |

| Method | Log/raw sign agreement | Spearman $\rho$ |
|---|---|---|
| MetaShift fixed | 95.2% | 0.829 |
| Standard SC | 92.5% | 0.868 |
| Nearest-neighbor DiD | 92.1% | 0.843 |

*Note.* Window sensitivity adds a full method-stability check for every target and donor window. Reporting-scale agreement is a concordance diagnostic and does not equate raw and log causal estimands.

**Table 20:** One-factor screening sensitivity with three required distinct donors.

| Setting | Eligible before donor rule | With at least 3 donors |
|---|---|---|
| Primary (100 km) | 563 | 238 |
| Coverage 70% | 563 | 243 |
| Coverage 80% | 563 | 232 |
| Stable window 45 days | 572 | 252 |
| Stable window 90 days | 543 | 214 |
| Transition gap 3 days | 512 | 212 |
| Transition gap 14 days | 589 | 254 |
| Donor radius 50 km | 563 | 102 |
| Donor radius 200 km | 563 | 426 |
| Correlation $\rho \geq 0.50$ | 563 | 241 |
| Correlation $\rho \geq 0.70$ | 563 | 230 |

*Note.* One predeclared setting changes at a time. These counts describe audit availability, not detected-effect frequency.

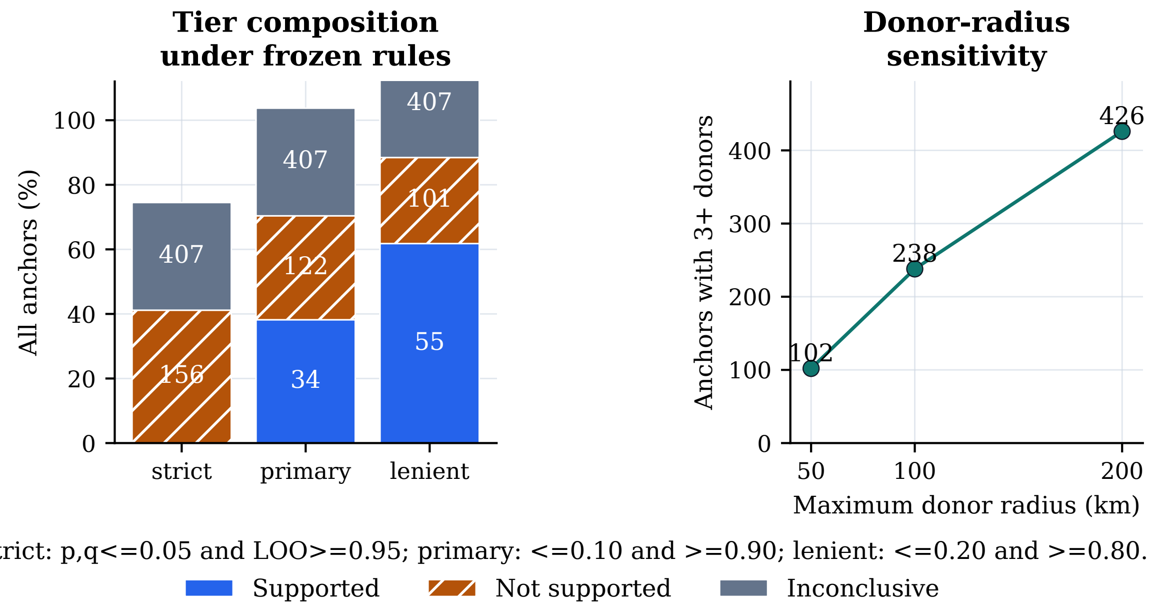


**Figure 18:** Predeclared evidence-tier and geographic-screening sensitivity. Changes describe availability and rule dependence, not physical causality or a search for a favorable specification.

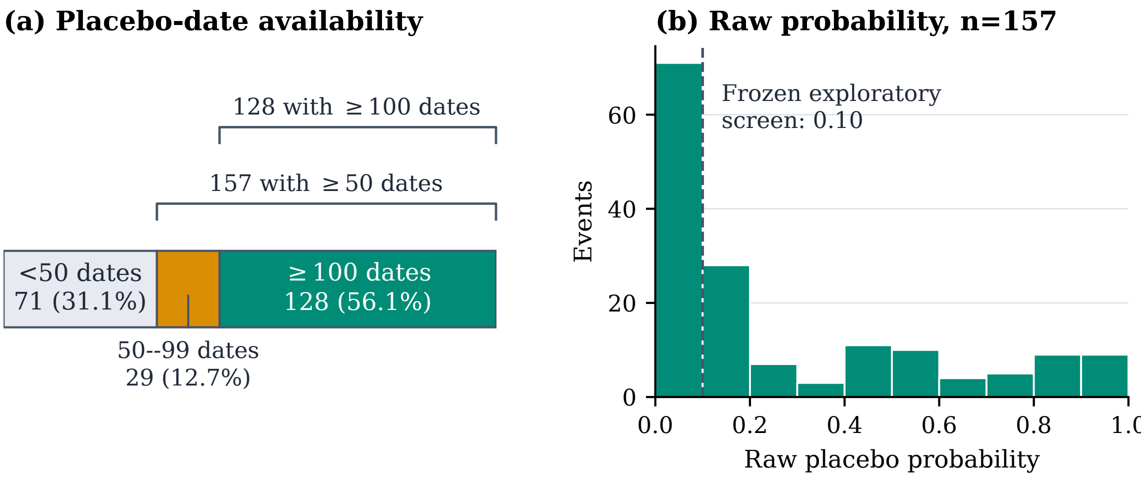


**Figure 19:** Supplementary time-placebo availability and raw within-event placebo-score distribution. These are diagnostic falsification quantities, not causal probabilities.

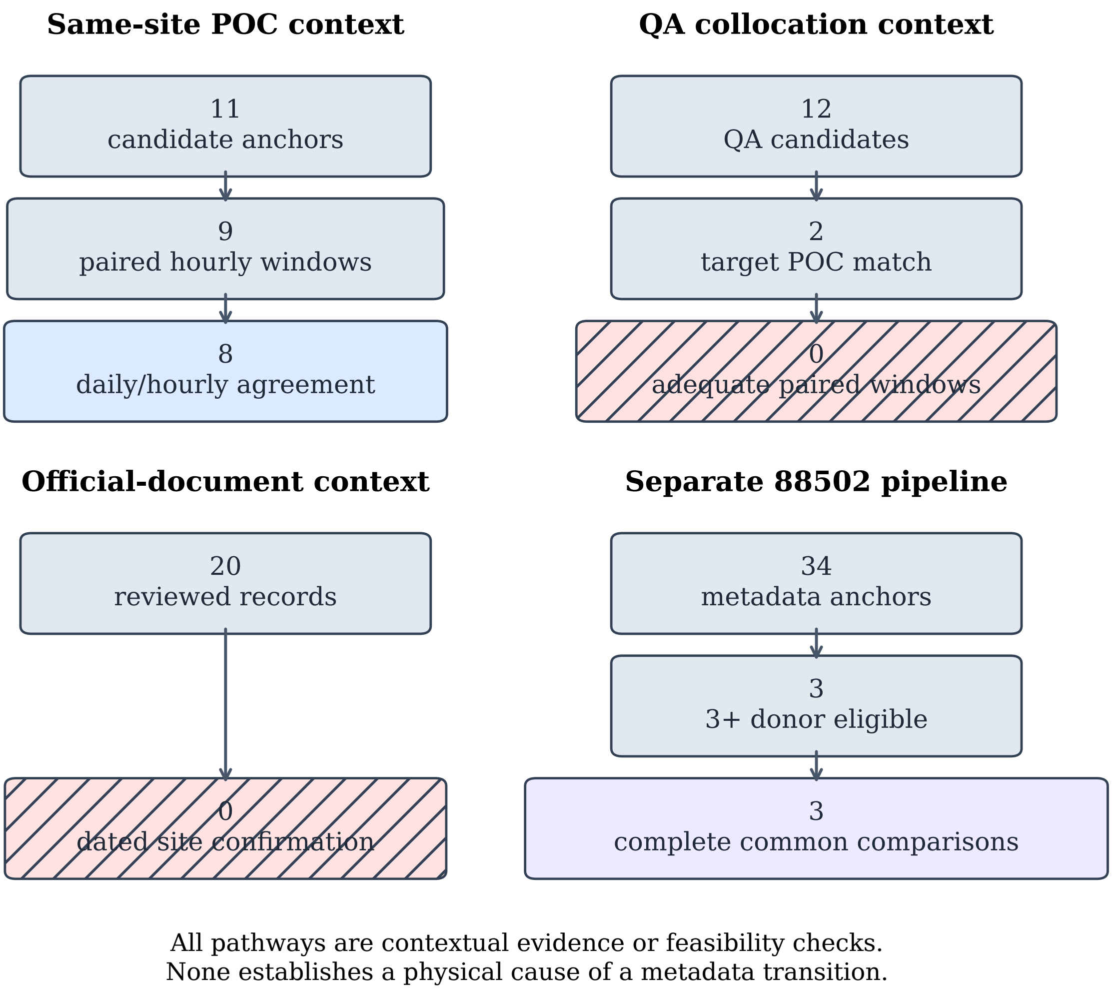


**Figure 20:** Supplementary contextual-evidence ladders for same-site POC, QA-collocation, targeted document review, and the separate 88502 pipeline. Unavailable evidence remains visible.

**Table 21:** Held-out synthetic interval coverage by method.

| Method | Fixed 95% coverage | Fixed mean width | Conformal 90% coverage | Conformal mean width |
|---|---|---|---|---|
| Standard SC | 67.3% | 0.265 | 98.8% | 0.396 |
| MetaShift fixed | 63.9% | 0.270 | 99.4% | 0.471 |
| MetaShift CV | 65.4% | 0.265 | 98.9% | 0.437 |
| Nearest-neighbor DiD | 66.7% | 0.287 | 99.6% | 0.565 |

*Note.* Each method has 3,200 evaluation effect instances. Fixed-weight intervals under-cover and split-conformal intervals over-cover under this frozen synthetic design.

**Table 22:** Deterministically selected representative audit cases.

| Case | Metadata anchor | Disposition | Saved diagnostic summary |
|---|---|---|---|
| Supported candidate | `12-057-0113 poc1-2023-12-01` | Complete | Effect -0.12777; fixed [-0.17258, -0.10511]; nested [-0.17399, -0.10560] |
| Not supported | `06-031-0004 poc8-2021-01-12` | Complete | Effect -0.08240; fixed [-0.21609, 0.03012]; nested [-0.24309, 0.03662] |
| Inconclusive | `01-103-0011 poc3-2023-02-01` | Donor insufficient | No counterfactual, effect, or interval is imputed. |

*Note.* Supported and not-supported cases minimize absolute distance to their within-tier median standardized score, with anchor ID tie-breaking. The inconclusive case is the lexicographically first donor-insufficient anchor and deliberately has no counterfactual or interval. Intervals are diagnostic, not calibrated physical-bias confidence intervals.

## A.4 Representative AQS cases

### A.4.1 Selection policy and source provenance

Examples can make an audit readable, but selected examples can also conceal a failure mode or overstate a method's scope. The three cases in this section are therefore not hand-picked for their visual contrast or effect magnitude. The display configuration selects one complete supported-candidate anchor and one complete not-supported anchor by minimum distance to the within-tier median standardized score, with anchor-ID tie-breaking. It selects the lexicographically first donor-insufficient inconclusive anchor to make the absence of a qualified counterfactual visible. The configuration, selected donors, archive manifest checksum, donor-table checksum, fitted weights, and reconstruction error are stored in the generated case-study manifest.

For the two complete cases, the renderer reads only the checksum-pinned public EPA archives, applies the frozen cleaning rule, reconstructs fixed-weight MetaShift with the reliability-prior penalty from pre-anchor data, and fails if the reconstructed saved log effect differs by more than $10^{-9}$. The raw archives remain outside version control. The third case intentionally has no reconstructed counterfactual: its audit disposition is that fewer than three qualified distinct physical donors were available.

### A.4.2 Interpretation boundaries

The supported-candidate panel illustrates a case in which the predeclared diagnostic conditions jointly hold. That status prioritizes the anchor for further station-history and technical-record review; it does not identify a device, a maintenance action, a calibration change, or a measurement bias. The not-supported panel illustrates the opposite distinction: a visible target series and a complete counterfactual are not sufficient when available diagnostics fail the frozen evidence rules. It would be misleading to discard that case merely because its residual plot is less dramatic.

The inconclusive panel is equally important. A target series may have an apparent before/after pattern, yet the protocol cannot form the common cross-site comparison if the distinct-donor

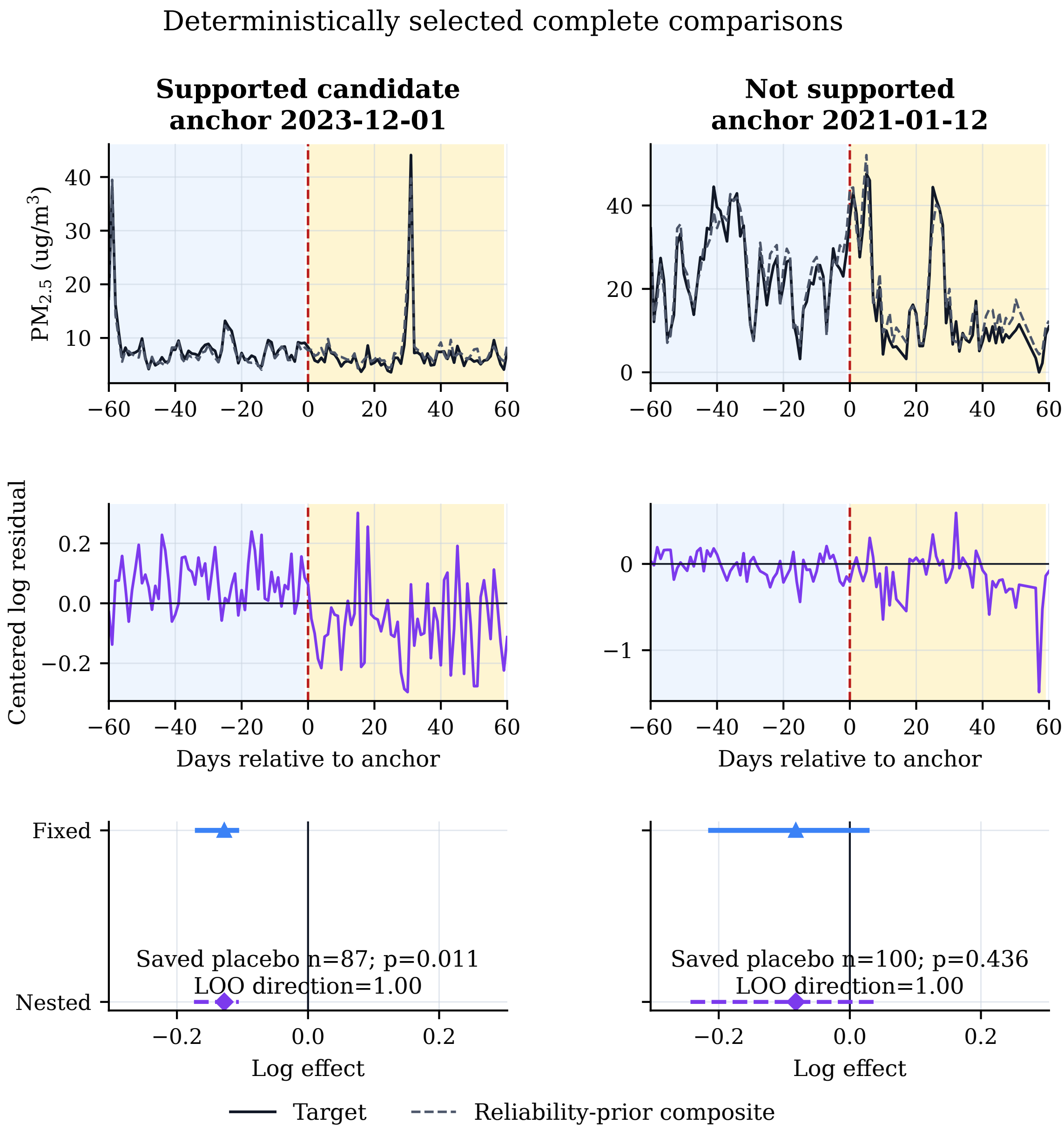


**Figure 21a:** Deterministically selected complete representative cases. The top row shows retained daily target values and the pre-anchor-fitted reliability-prior MetaShift counterfactual in $\mu$g/m$^3$; the middle row shows calibrated log residuals; and the lower row shows saved placebo summaries with fixed and nested diagnostic intervals. Blue and amber shading mark the 60-day pre- and post-anchor comparison windows. The figure displays saved diagnostic outputs, not evidence of a physical instrument event.

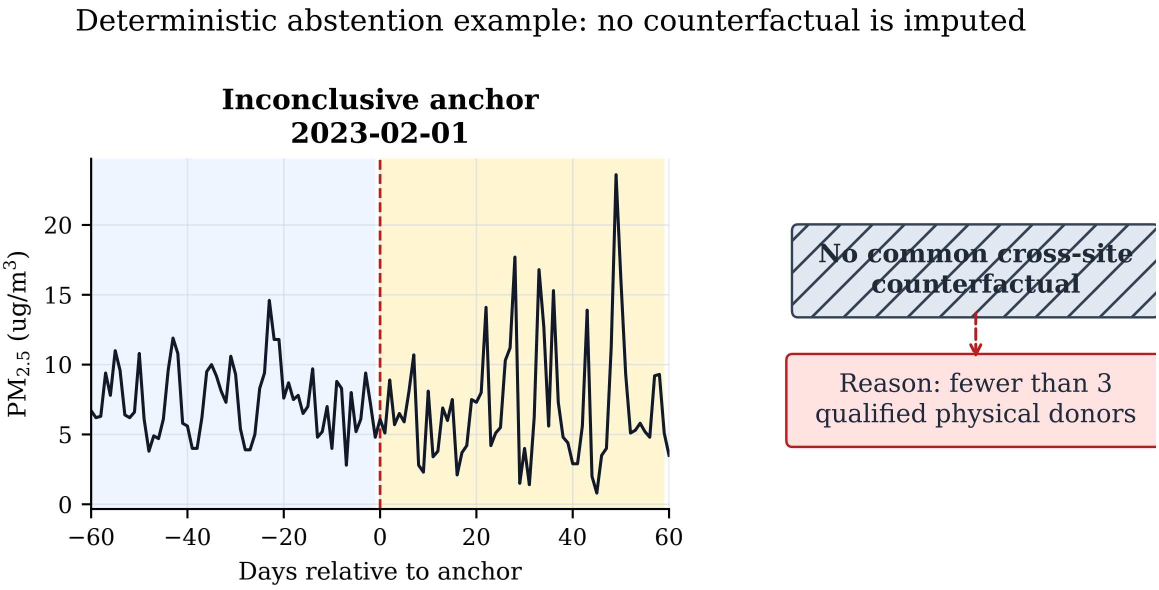


**Figure 21b:** Deterministic representative audit abstention. The retained target series is shown without a constructed counterfactual because fewer than three qualified distinct physical donors were available. The protocol therefore records an unavailable common comparison and does not impute a placebo, effect, or interval.

requirement fails. MetaShift-Bench therefore abstains rather than borrowing a same-site POC, relaxing the donor rule after inspection, or assigning a physical explanation. This case makes the audit's coverage boundary concrete: lack of a qualified comparison is lack of evidence for this protocol, not evidence that nothing happened.

## A.5 Applicability and anchor concentration

The frozen inventory has 393 of 563 anchors (69.8%) dated in 2023. Two reported code-pair transitions, 236 to 636 and 238 to 638, account for 354 of these 393 anchors (90.1%). This concentration is descriptive only: a code pair, time cluster, or method-context document does not establish an administrative, software, hardware, policy, calibration, or data-processing cause for an individual site record.

**Table 23:** Appendix-only temporal concentration of reported 88101 metadata anchors.

| Old code | New code | Abbreviated reported metadata change | Count |
|---|---|---|---|
| 236 | 636 | T640, baseline reported name -> T640, alignment-enabled | 234 |
| 238 | 638 | T640X, baseline reported name -> T640X, alignment-enabled | 120 |

*Note.* Of 393 2023 anchors, these two pairs account for 354 (90.1%). The largest date is 2023-07-01 (50), and the largest state-code subtotal is 42 (42). Descriptions abbreviate reported AQS Method Name strings; they do not establish a cause.

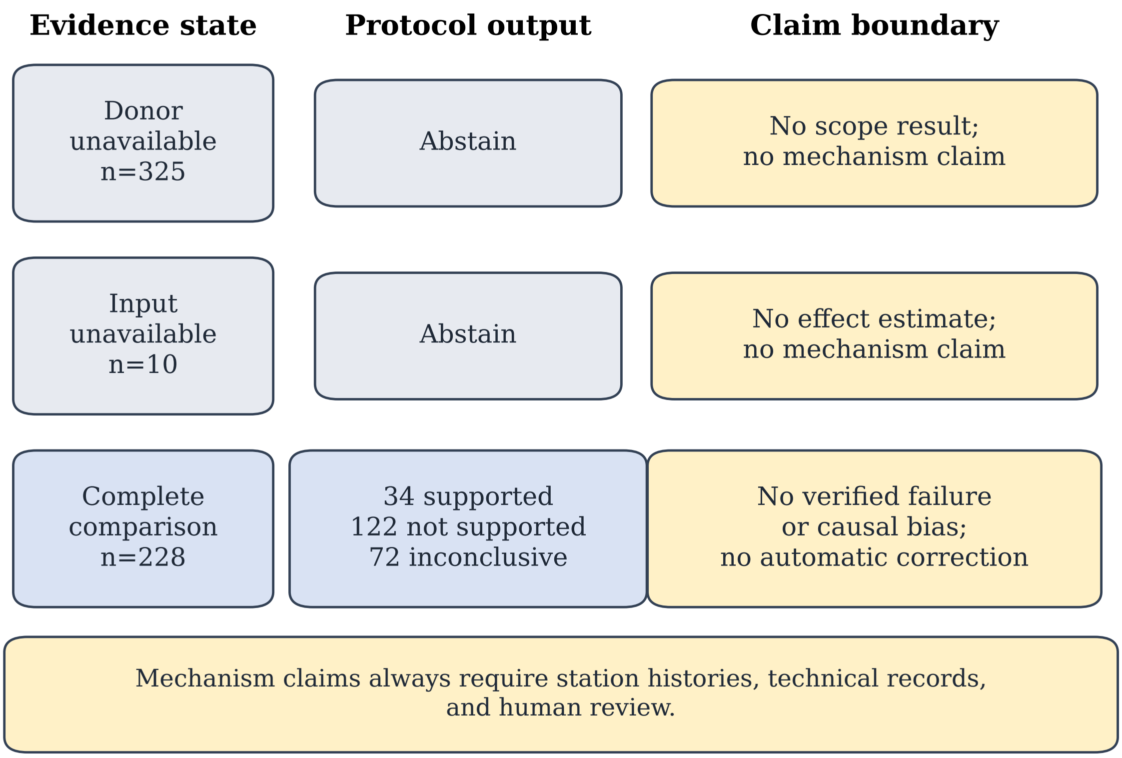


**Figure 22:** Claim-boundary matrix for the AQS audit. Rows partition all 563 primary anchors into donor-insufficient, input-window-failure, and complete-comparison states; the complete row separately reports observational evidence tiers. Every row excludes a physical-mechanism conclusion.

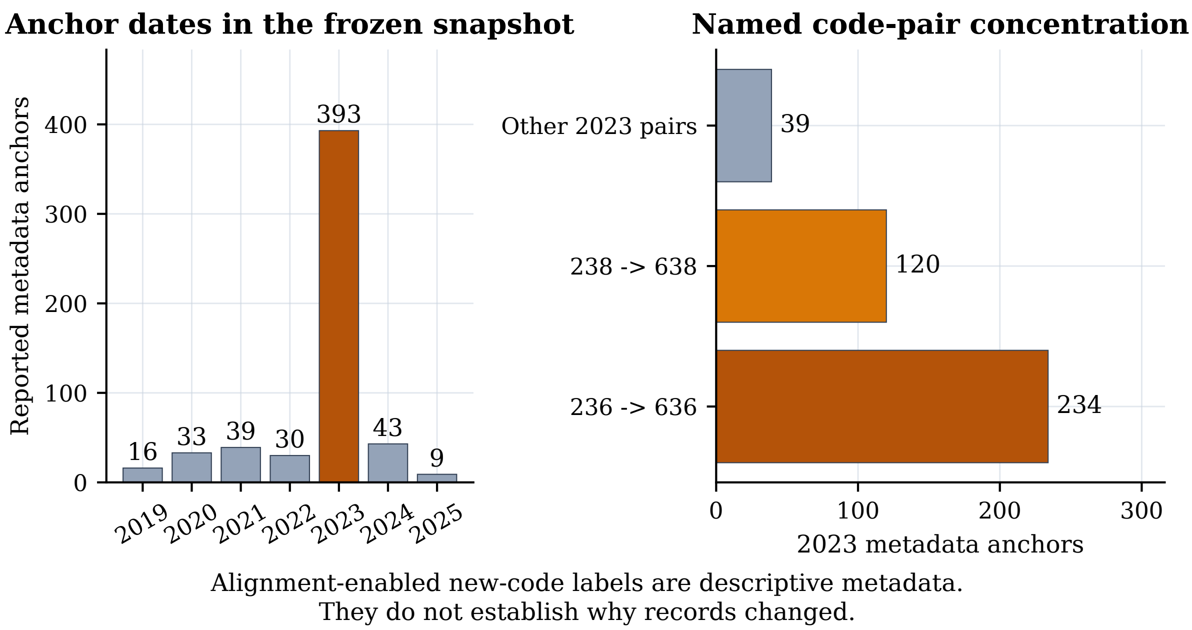


**Figure 23:** Descriptive concentration of reported 88101 metadata anchors. The association between code-pair labels and dates is not evidence of cause.

## A.6 Claim classification and threats to validity

The following tables distinguish established theory, construction-specific derivations, finite synthetic observations, and claims that require additional external evidence.

**Table 24:** Claim boundaries imposed by the MetaShift-Bench protocol.

| The evidence can support | The evidence does not establish | Required interpretation |
|---|---|---|
| A reproducible metadata-anchored local residual audit | That a Method Code transition is a physical replacement or fault | Treat the transition as an anchor, not physical ground truth. |
| Benchmark-specific synthetic trade-offs among frozen methods | A robust aggregate MetaShift superiority claim | Report the paired intervals and retain the negative result. |
| Protocol-defined evidence tiers and abstentions | Instrument bias, true pollution correction, or causal attribution | Use tiers only to prioritize further records-based review. |
| Conditional interval behavior on frozen synthetic data | Coverage-calibrated confidence intervals for real-event physical bias | Describe real-event intervals as diagnostic. |

*Note.* These boundaries are design constraints, not post hoc caveats.

**Table 25:** Classification and boundary of the principal claims.

| Statement | Status | Boundary |
|---|---|---|
| Information refinement and risk–coverage principles | Established theory | Cited background, not a MetaShift-Bench invention. |
| Target-only impossibility under the stipulated paired law | Established theory specialized to this construction | Requires balanced matched target laws and no target-side leakage. |
| Availability-normalized partial-scope identity and interval separation | New construction-specific derivation | Requires analysis-scale intervention and stated finite envelopes. |
| Held-out frontiers, gain, and certificate behavior | Frozen empirical observation | Applies only to the finite predeclared policies and generator. |
| Real AQS scope accuracy or physical mechanism | Explicit nonclaim | Requires independent labels, records, and human review. |

**Table 26:** Threats to validity, current boundary, and needed future evidence.

| Validity class | Current boundary | Needed future evidence |
|---|---|---|
| Construct | Method Code and POC records are anchors, not physical device, calibration, siting, or maintenance ground truth. | Dated station histories, calibration records, deployment logs, and independently reviewed taxonomy. |
| Synthetic scope | Target-fixed labels and bounded panels may not represent operational networks; certificate inputs are known only to the generator. | Independent blind generators, naturally occurring labeled scope events, and admissible external records. |
| Internal | Weather, emissions, processing, and donor instability can make a target-reference residual; diagnostics constrain but do not identify cause. | Meteorological covariates, emissions records, network operations logs, and prospective comparator studies. |
| Statistical | Selection, serial dependence, multiplicity, and incomplete resampling limit real-event interval interpretation. | A computationally feasible predeclared full selection-aware coverage study on a new blind set. |
| External | The U.S. $PM_{2.5}$ archive, availability rules, radius, and intervention families limit transfer. | Independent footprint-disjoint benchmarks from other networks, eras, and pollutant parameters. |

# B Reproducibility package

The project preserves two immutable evidence streams. The stable-regime benchmark and AQS deployment audit are bound to:

`v0.3.2-evidence-final`
`57d678ecabebff724d898abe626c9ef80538775b`

Their manifest records the source artifacts, checksums, and generated presentation assets. The target-fixed answerability experiment is separately bound to:

`v0.5.0-answerability-freeze`
`14fd0fee4fb015e6c661299041e35ff704a27286`
receipt SHA-256 prefix `954fc9b56a8f5266...`

Its result verifier checks source and protocol bindings, full-grid accounting, target and $q = 0$ identity, split separation, and deterministic result consistency without rerunning the executor. The execution authority is recorded under `v0.5.0-answerability-execution-claim`.

The repository distinguishes archived-result verification from third-party reproduction. Read-only verification confirms that the preserved artifacts and their claimed provenance match. Reproduction on another environment requires the pinned requirements and records comparison; the frozen two-environment check applies to the designated core artifacts, not every local file, timestamp, cache, or renderer output.

The report build regenerates presentation-only tables and figures from verified inputs, validates source structure, references, numeric claims, and asset determinism, then performs a clean LaTeX build, font audit, and visual review. It never changes the archived evidence. The one-time target-fixed executor is not part of the document-build path and must not be rerun to improve or replace its results.

**Table 27:** Frozen evidence release verification record.

| Record | Value |
|---|---:|
| Evidence tag | v0.3.2-evidence-final |
| Frozen evidence commit | `57d678ecabeb` |
| Synthetic result label | stable_full_v2 |
| Case manifest SHA-256 | `065b1b65c231c529...` |
| Release-gate checks passed | 35/35 |
| Document-consistency checks passed | 12/12 |
| Markdown number checks passed | 57/57 |
| Two-environment core hashes matched | 34 |

*Note.* The two-environment statement applies only to the listed designated core artifacts in the frozen reproducibility comparison, not to every local file.

# C Acknowledgements, contribution statement, and required disclosures

**HUMAN COMPLETION REQUIRED.** This section is an intentionally uncompleted truthful-disclosure template. It must be replaced by a 500–1500 word account, verified by every student author and supervising teacher, before the report is submitted. Do not state that a person performed work they did not perform. Do not state that an advisor, school, organization, or AI system did or did not contribute unless the authors have verified the statement. Before submission, all student authors and supervising teachers must verify the exact final PDF and complete the required academic-integrity declaration, signatures, institutional stamps, and final truthfulness attestation.

### Topic origin

**HUMAN COMPLETION REQUIRED:** Explain how the team selected the question, including the intellectual path from the original observation or reading to the metadata-anchor audit formulation. State whether a supervising teacher or other person suggested the topic and how the students refined the question.

### Data acquisition and analysis

**HUMAN COMPLETION REQUIRED:** Identify who downloaded or reconstructed the public EPA AirData inputs, who inspected data rules, who implemented or reviewed the cleaning and donor-uniqueness logic, and who ran or checked the frozen analyses. State only the actual use of public AQS bulk files, public API endpoints, and local credentials. Do not include credentials, raw API responses, or private information.

### Experimental design and writing

**HUMAN COMPLETION REQUIRED:** State which student(s) designed the synthetic evaluation, maintained the no-tuning boundary, generated figures and tables, interpreted limitations, and drafted or revised each report section. Describe the physical-donor uniqueness defect and remediation only if the authors personally understand and verify that account.

### Advisor relationship, compensation, and external help

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### AI-use disclosure

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